

Healthcare Cyber-Physical Systems (H-CPS)

Keynote – National Symposium on "Transforming Healthcare Through AI, Biotechnology, Digital Health & Smart Medical Technologies" (AIMED 2026)

<https://www.ganpatuniversity.ac.in/digitalhealthtech/digitalhealthtech-home>



Homepage:
www.smohanty.org

Ahmedabad, India, 7--8 Aug 2026

Prof./Dr. Saraju Mohanty
University of North Texas, USA.

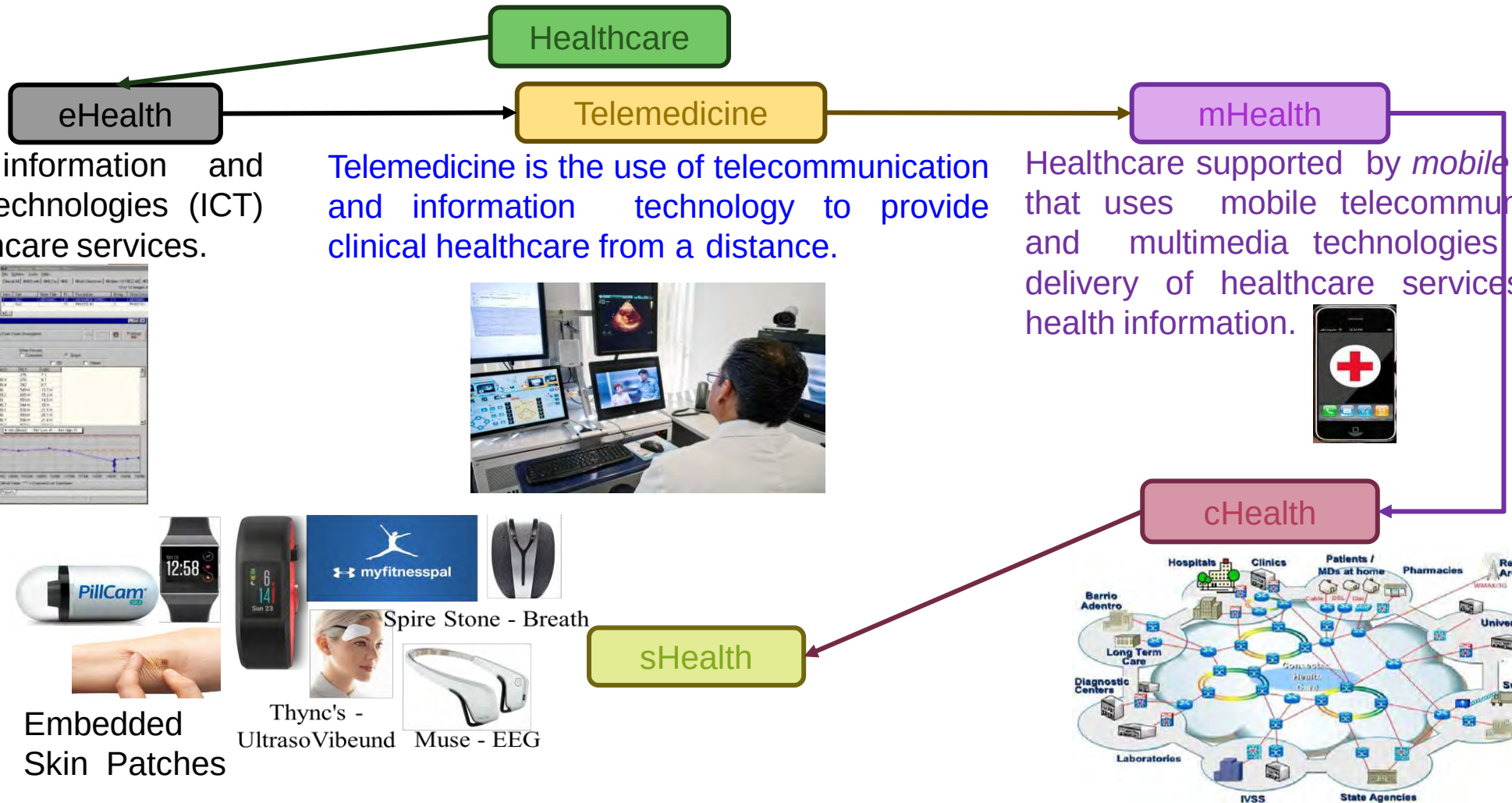


Outline

- Smart Healthcare – Introduction
- Smart Healthcare – Challenges Against Sustainability
- Selected Cybersecurity Solutions for IoT/CPS
- Drawbacks of Existing Cybersecurity Solutions of IoMT/H-CPS
- Security by Design (SbD) Principle
- Security by Design (SbD) Example Solutions
- Trustworthy Pharmaceutical Supply Chain
- Trustworthy Medical Prescription
- Conclusion and Future Directions

Smart Healthcare – Introduction

Healthcare → Smart Healthcare



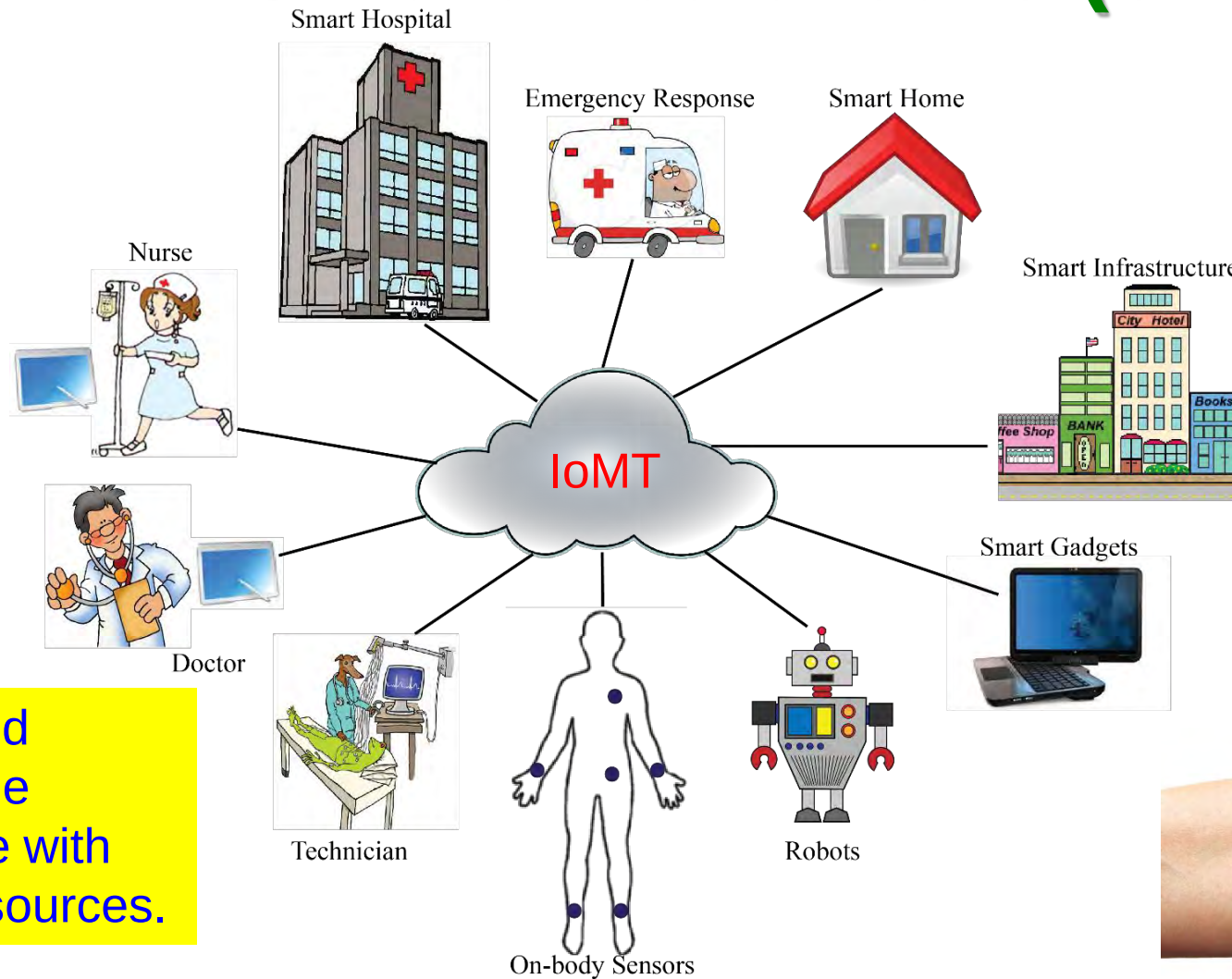
The use of information and communication technologies (ICT) to improve healthcare services.

Telemedicine is the use of telecommunication and information technology to provide clinical healthcare from a distance.

Healthcare supported by *mobile devices* that uses mobile telecommunications and multimedia technologies for the delivery of healthcare services and health information.

Source: **S. P. Mohanty**, "Smart Healthcare: From Healthcare to Smart Healthcare", ICCE 2020 Panel, Jan 2020.

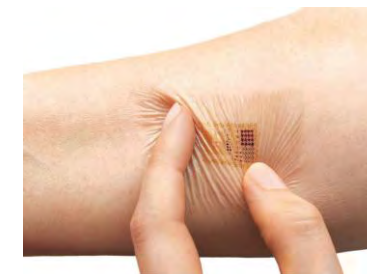
Smart Healthcare (sHealth)



Fitness Trackers



Headband with Embedded Neurosensors

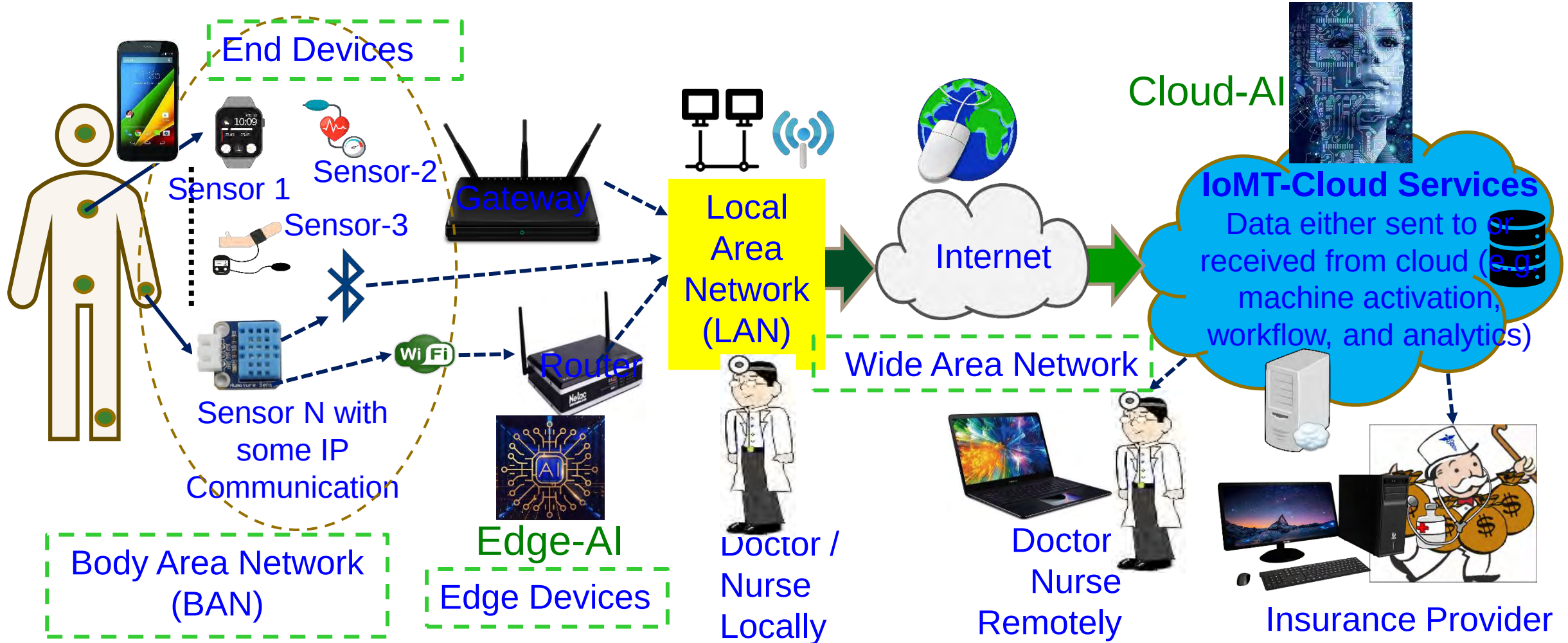


Embedded Skin Patches

Quality and sustainable healthcare with limited resources.

Source: P. Sundaravadivel, E. Kougianos, S. P. Mohanty, and M. Ganapathiraju, "Everything You Wanted to Know about Smart Health Care", *IEEE Consumer Electronics Magazine (MCE)*, Vol. 7, Issue 1, January 2018, pp. 18-28.

Smart Healthcare – Healthcare CPS

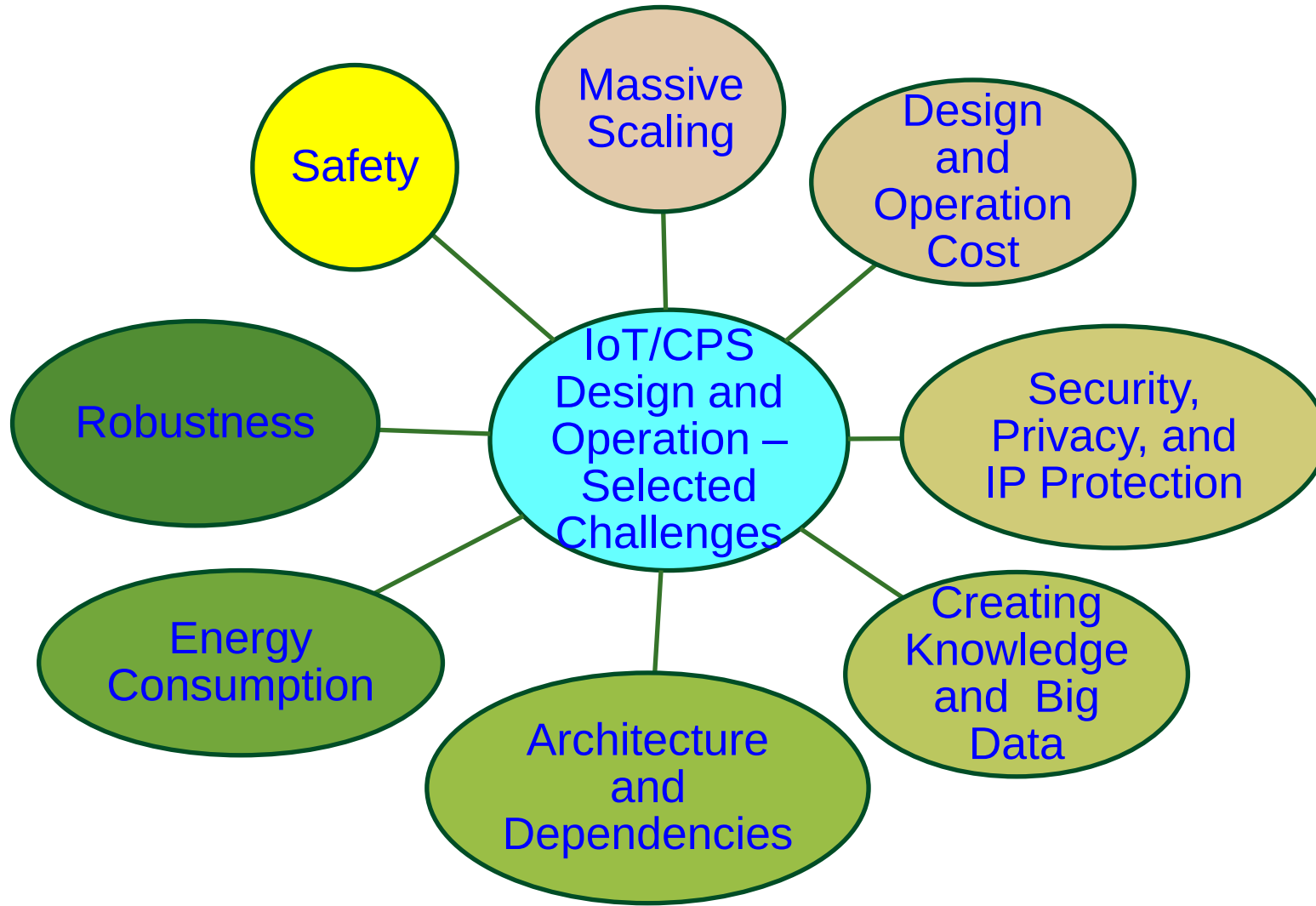


Frost and Sullivan predicts smart healthcare market value to reach US\$348.5 billion by 2025.

Source: S. P. Mohanty, Secure IoT by Design, Keynote, 4th IFIP International Internet of Things Conference (IFIP-IoT), 2021, Amsterdam, Netherlands, 5th November 2021.

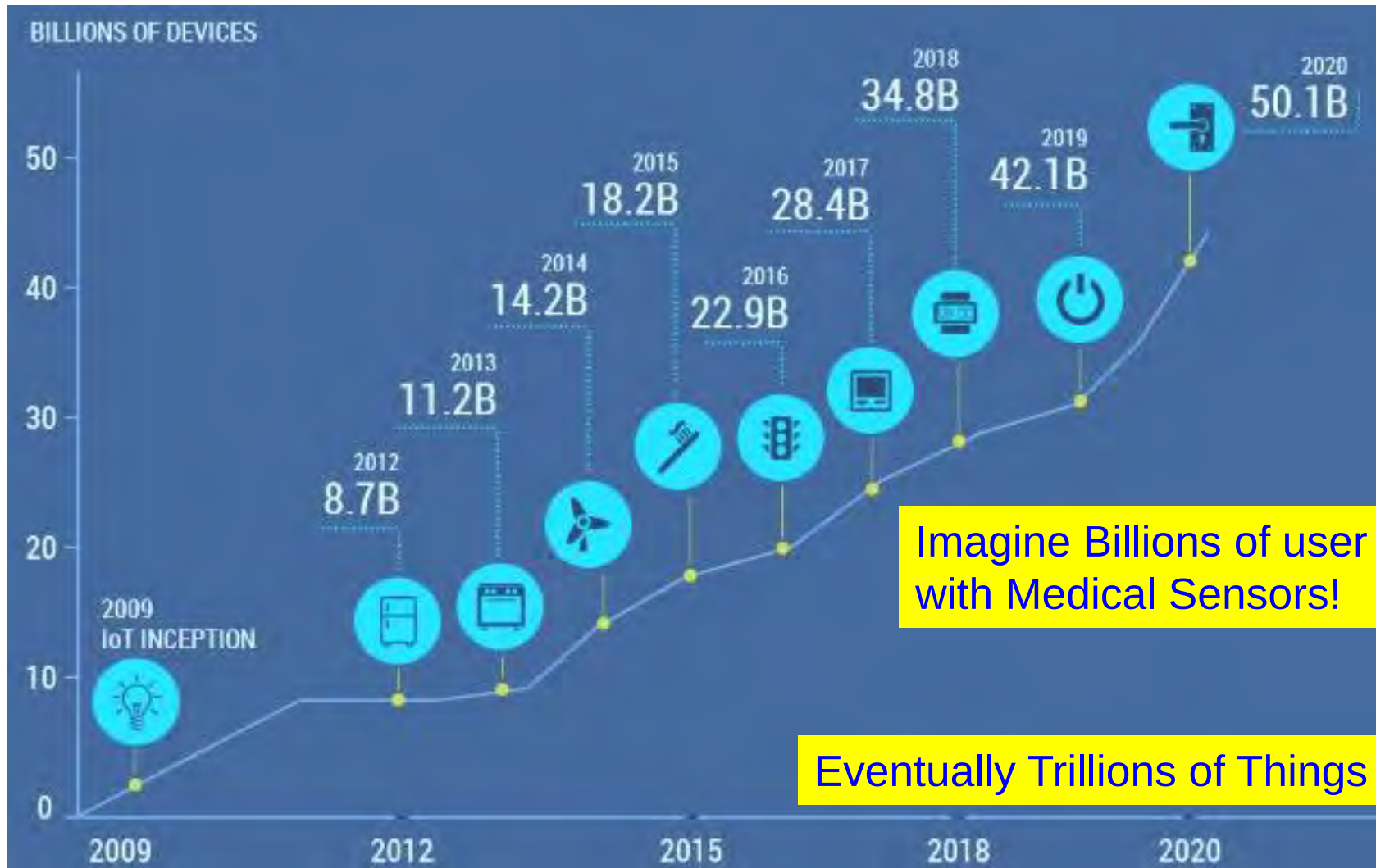
Smart Healthcare – Challenges Against Sustainability

CPS – Sustainability Challenges

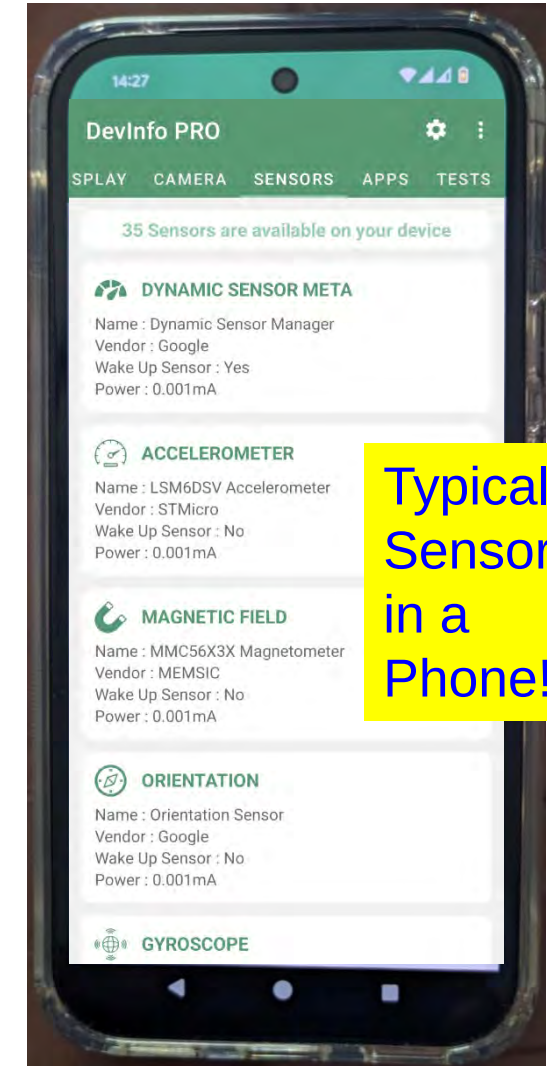


Source: Mohanty ICIT 2017 Keynote

Massive Growth of Sensors/Things



Source: <https://www.linkedin.com/pulse/history-iot-industrial-internet-sensors-data-lakes-0-downtime>



Challenges of Data in IoT/CPS are Multifold

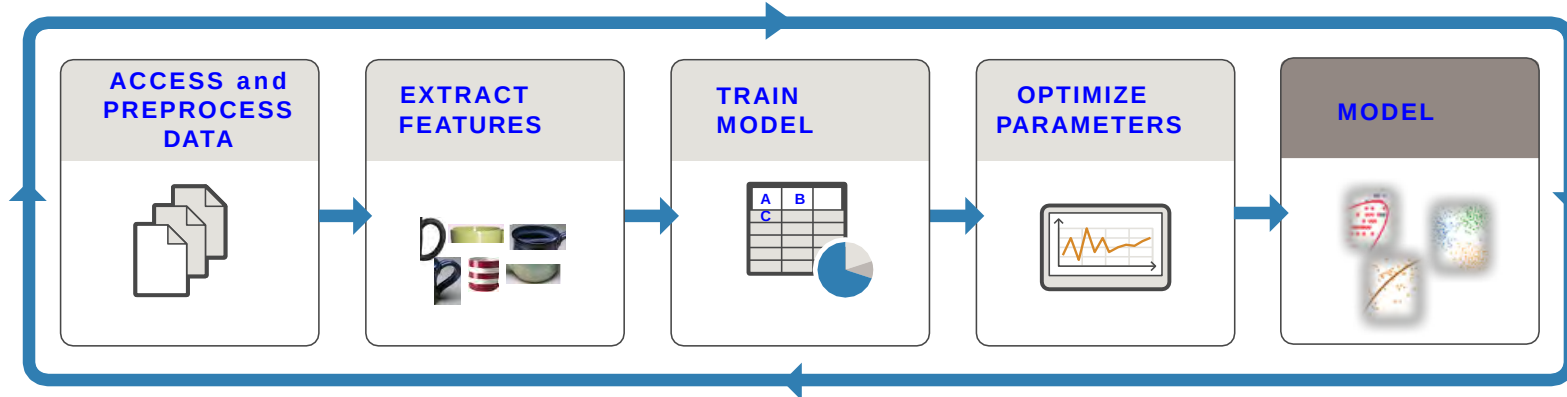


Deep Neural Network (DNN) - Resource and Energy Costs

TRAIN: Iterate until you achieve satisfactory performance.

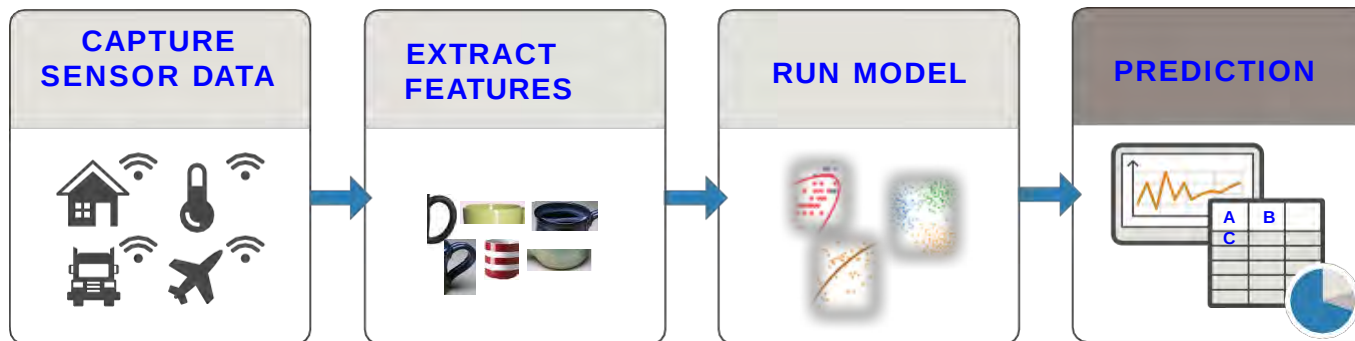
Needs Significant:

- Computational Resource
- Computation Energy



PREDICT: Integrate trained models into applications.

Limited Computational Capability
Limited Battery Life



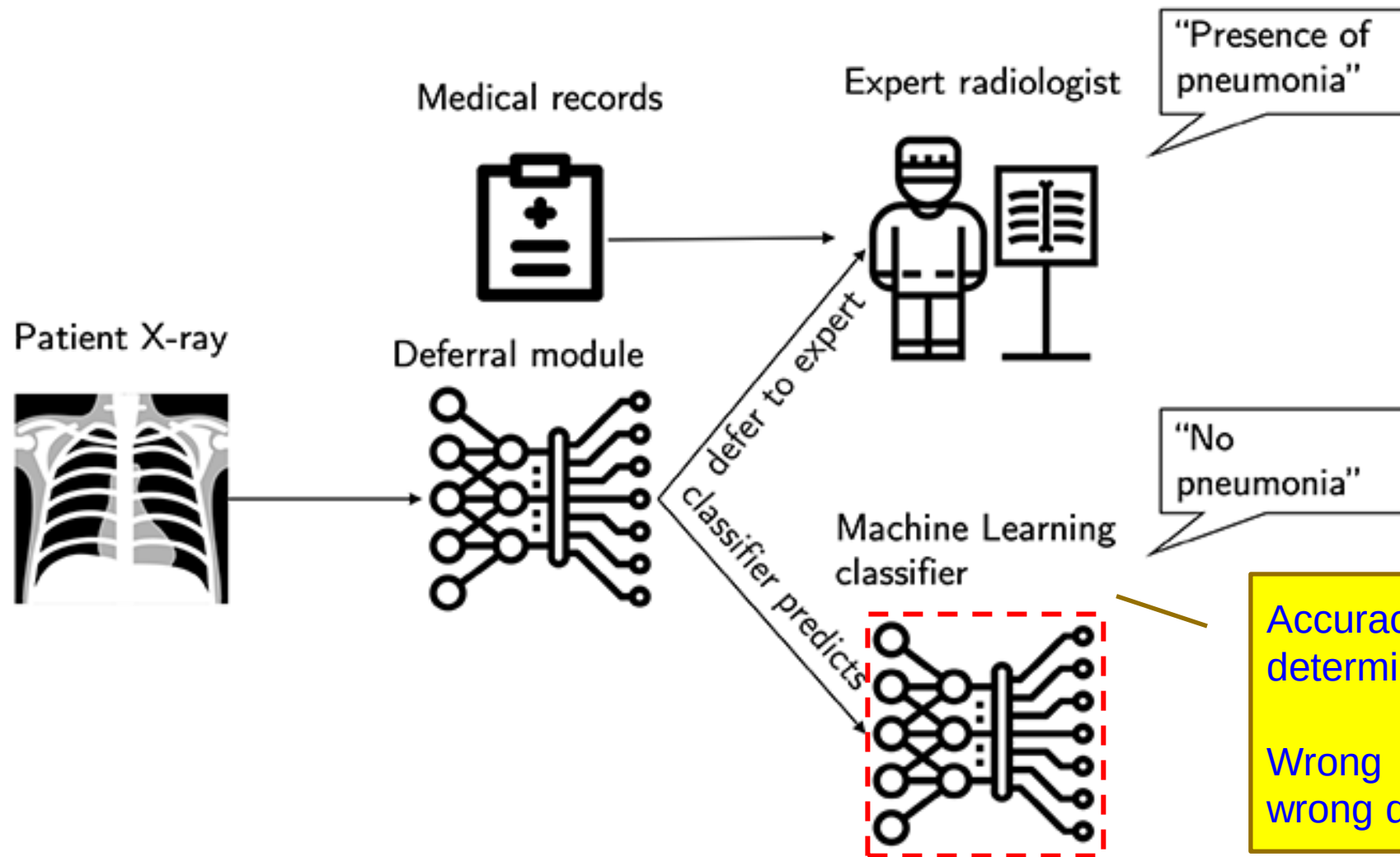
Needs:

- Computational Resource
- Computation Energy



Source: <https://www.mathworks.com/campaigns/offers/mastering-machine-learning-with-matlab.html>

Wrong ML Model → Wrong Diagnosis



Source: <https://www.healthcareitnews.com/news/new-ai-diagnostic-tool-knows-when-defer-human-mit-researchers-say>

Smart Healthcare - Cybersecurity and Privacy Issue

Selected Smart Healthcare Security/Privacy Challenges

Data Eavesdropping

Data Confidentiality

Data Privacy

Location Privacy

Identity Threats

Access Control

Unique Identification

Data Integrity

Device Security

Impersonation Attacks

Eavesdropping Attacks

Smart Healthcare



Reverse Engineering Attacks

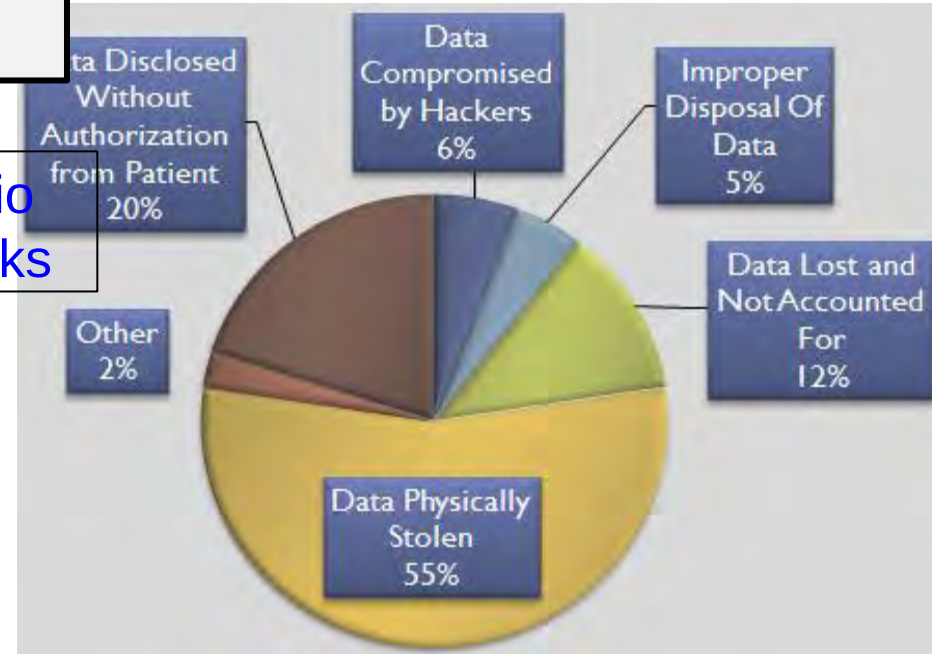
Radio Attacks



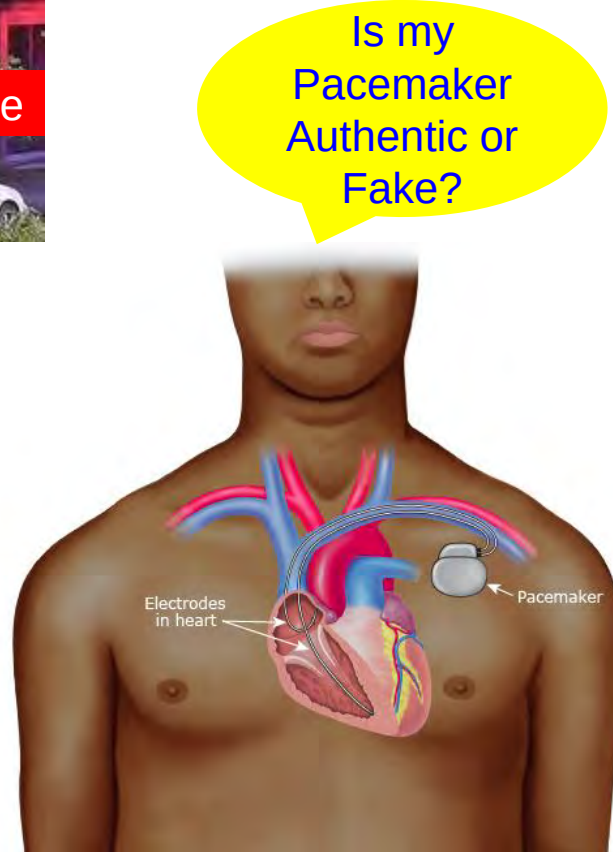
HIPAA

Health Insurance Portability and Accountability Act

HIPPA Privacy Violation by Types



Fake is Cheap – Why not Buy?



Fake Data and Fake Hardware – Both are Equally Dangerous in CPS



AI can be fooled by fake data



AI can create fake data (Deepfake)



Authentic
An implantable medical device



Authentic



Fake

A plug-in for car-engine computers

Counterfeits in Healthcare



The **original** product:

- sold in a white box with blue borders
- contains sixty (60) 500mg tablets
- divided on four (4) silver blister packs, each containing fifteen (15) tablets

The **fake** product:

- sold in a white box with no border
- contains sixty (60) 500mg tablets
- divided on six (6) silver with blue blister packs, each containing ten (10) tablets

Source: GA-FDD (Government Analyst – Food and Drug Department) issues warning over “fake” drug on local market,
<https://www.inewsguyana.com/ga-fdd-issues-warning-over-fake-drug-on-local-market/>

Daflon 500 is used to treat gravitational (stasis) dermatitis and dermatofibrosclerosis

E-Prescription is the Need of the Hour

Prescription Drug Type	Annual Abusers	% Among Rx Abusers	% Among Americans
Painkillers	9.7 million	59.5%	3.43%
Opioids Alone	9.3 million	57.1%	3.29%
Sedatives	5.9 million	36.2%	2.08%
Stimulants	4.9 million	30.1%	1.73%
Benzodiazepine Alone	4.8 million	29.4%	1.70%
All Prescription Drugs	16.3 million	100%	5.76%

Reduced Fraud and Abuse

Blockchain Immutability
Combats prescription fraud and abuse

Enhanced Security and Privacy:

Provides security and integrity of the medical data

Efficiency and Accuracy

Accuracy can be improved to reduce medication errors

Interoperability

Seamless data exchange between healthcare providers

Addressing Opioid Crisis

Prevents misuse and abuse of opioids

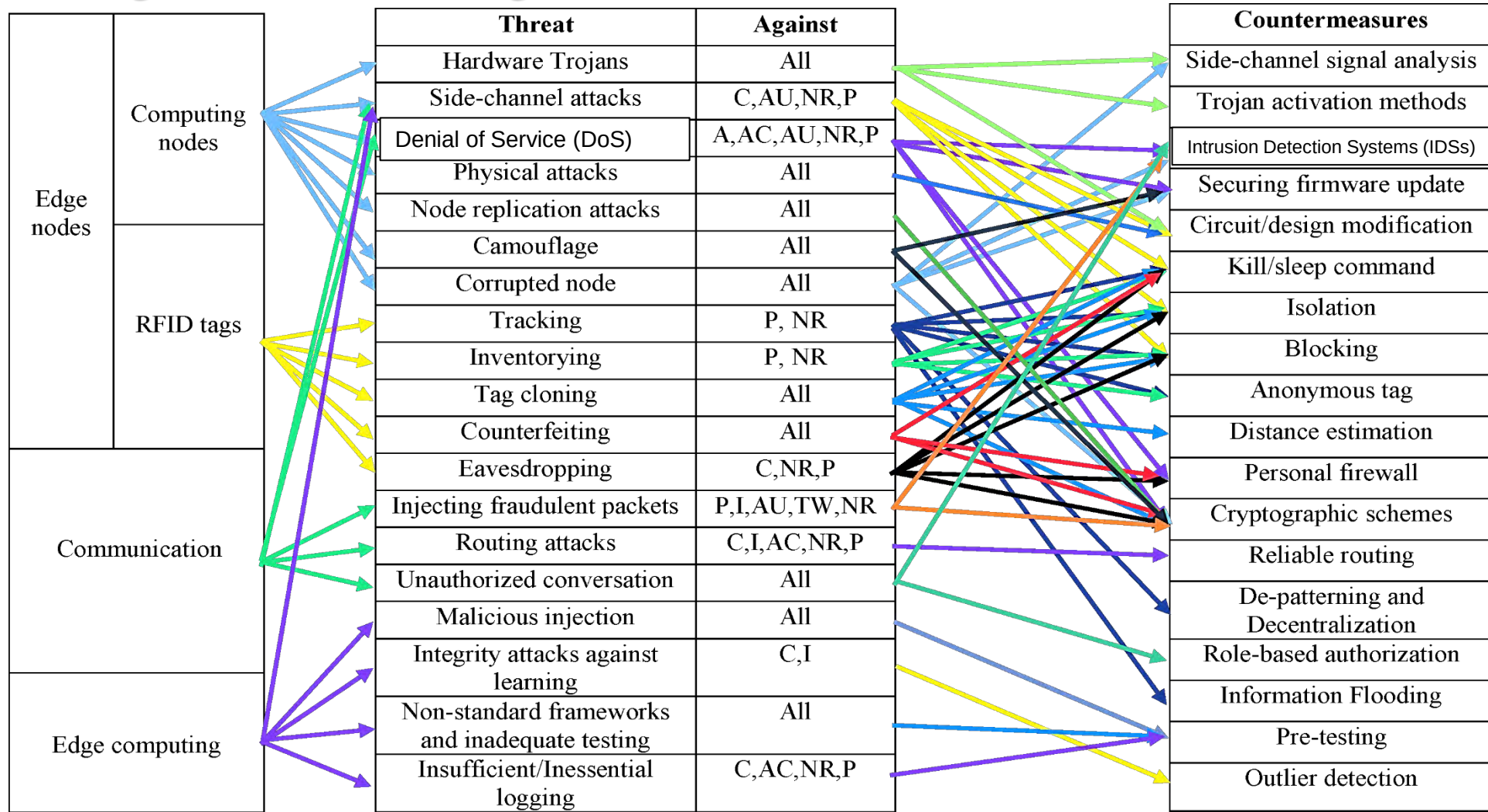
- 16M – 6% of Americans over the age of 12 abuse prescriptions in a year.
- 2M – 12% of prescription drug abusers are addicted.

Statistics Source: <https://drugabusestatistics.org/prescription-drug-abuse-statistics/>

Cybrsecurity Solutions for IoT/CPS



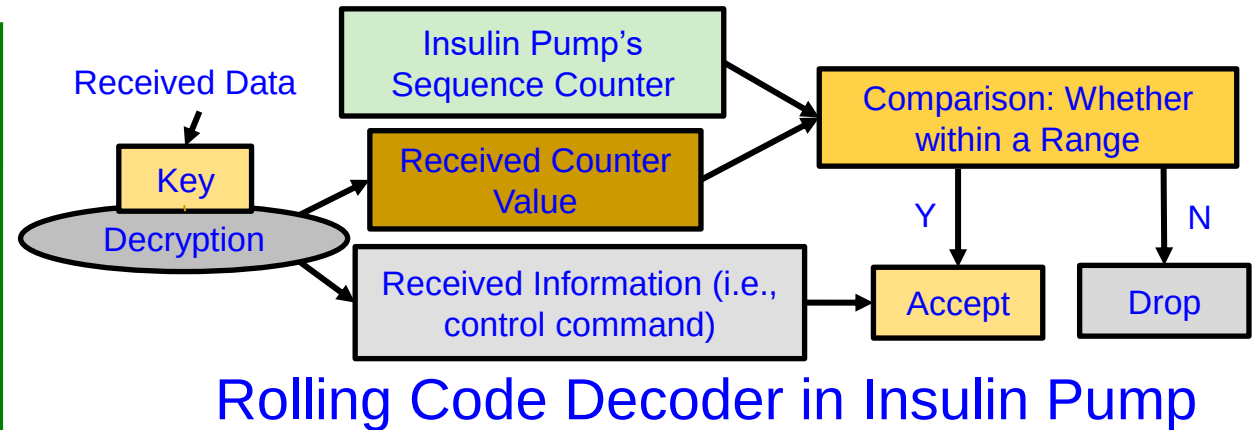
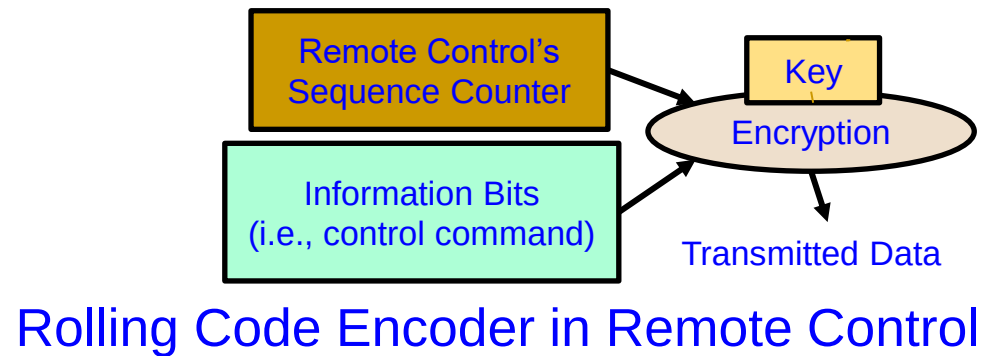
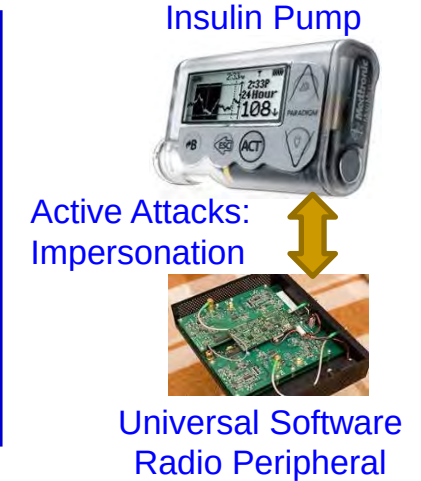
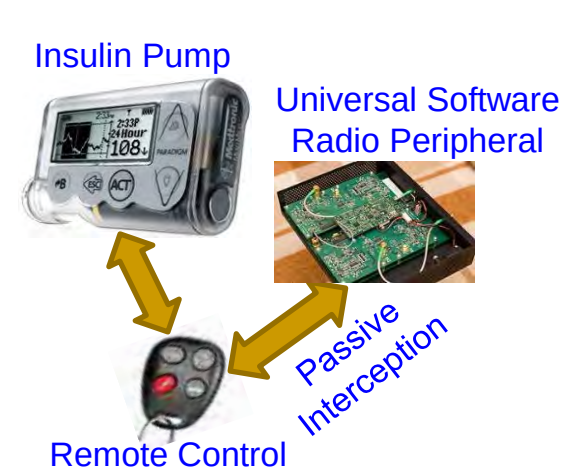
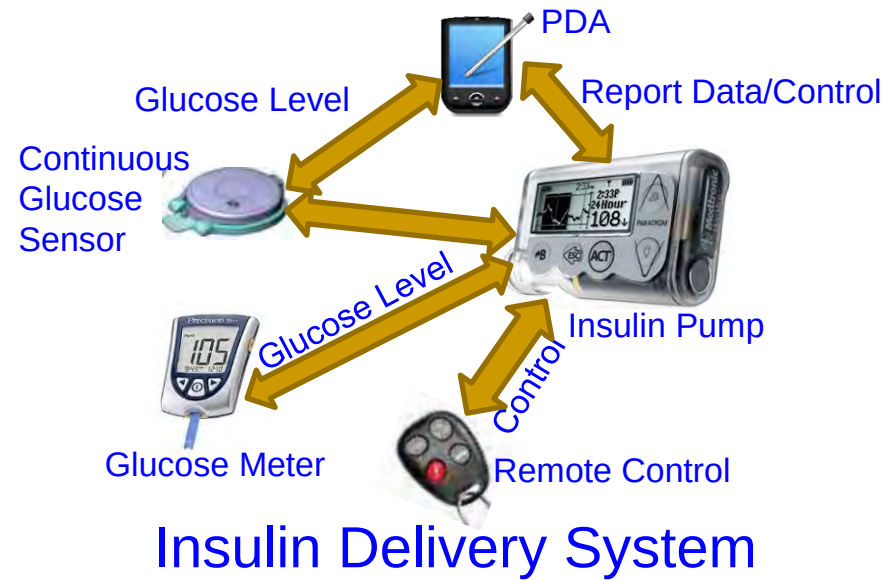
IoT Cybersecurity - Attacks and Countermeasures



C- Confidentiality, I – Integrity, A - Availability, AC – Accountability, AU – Auditability, TW – Trustworthiness, NR - Non-repudiation, P - Privacy

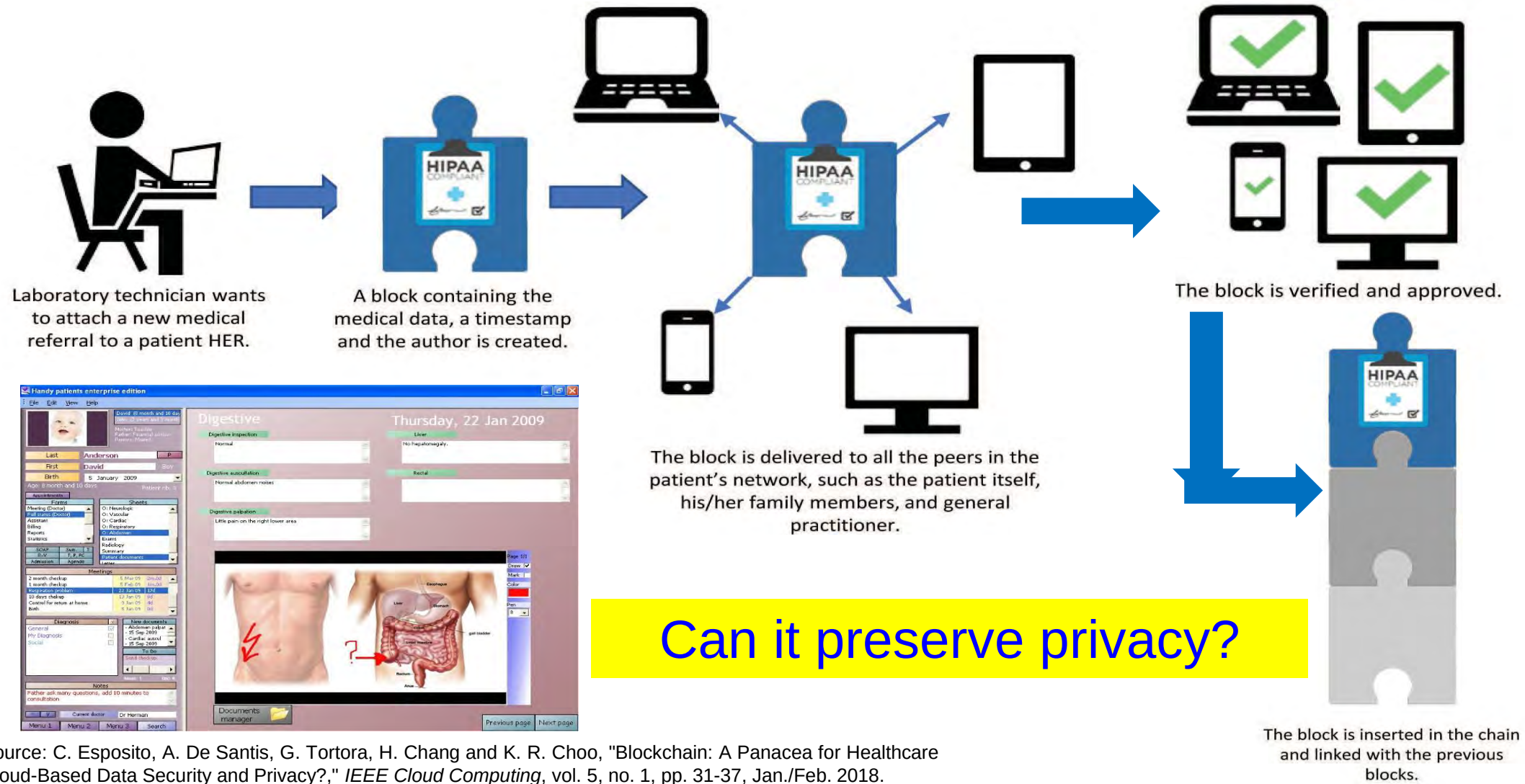
Source: A. Mosenia, and Niraj K. Jha. "A Comprehensive Study of Security of Internet-of-Things", *IEEE Transactions on Emerging Topics in Computing*, 5(4), 2016, pp. 586-602.

Smart Healthcare Cybersecurity

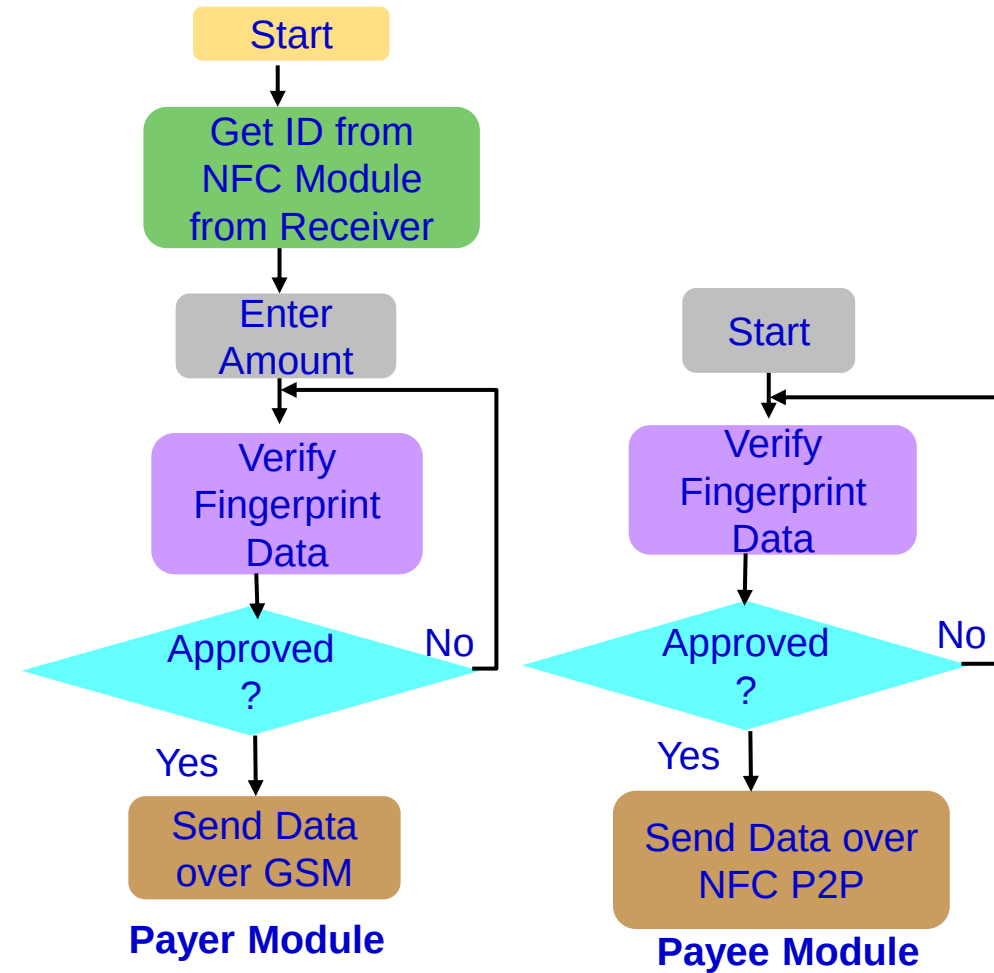
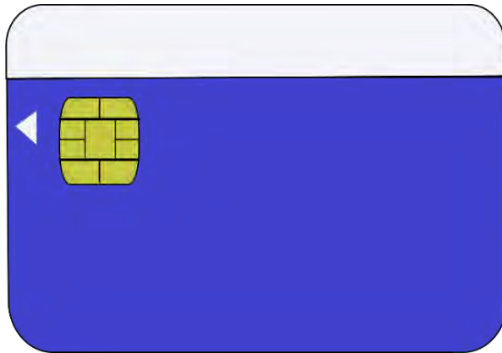
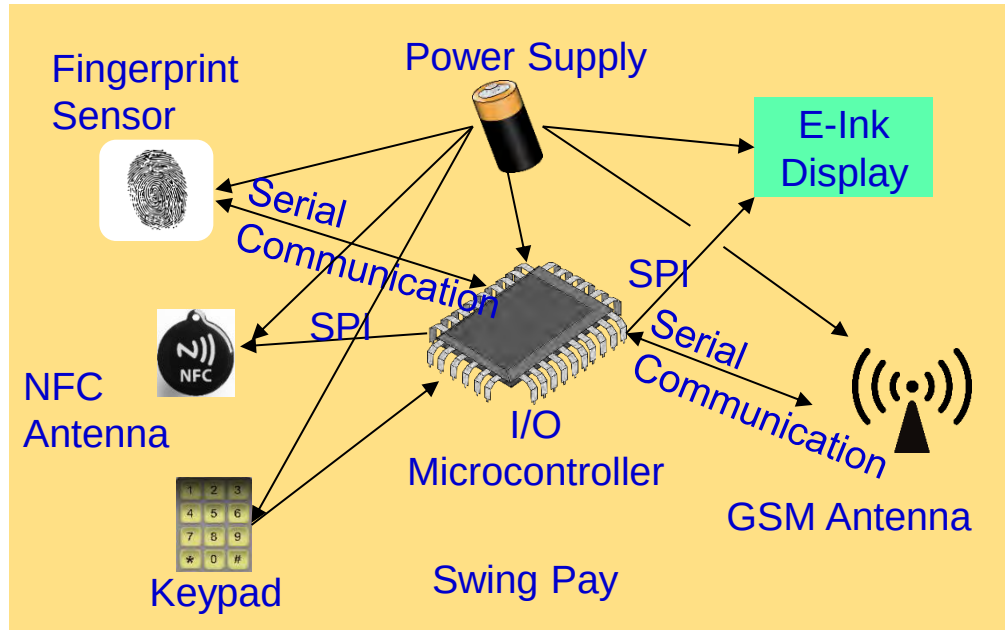


Source: Li and Jha 2011: HEALTH 2011

Blockchain in Smart Healthcare



Our Swing-Pay - NFC Cybersecurity Solution



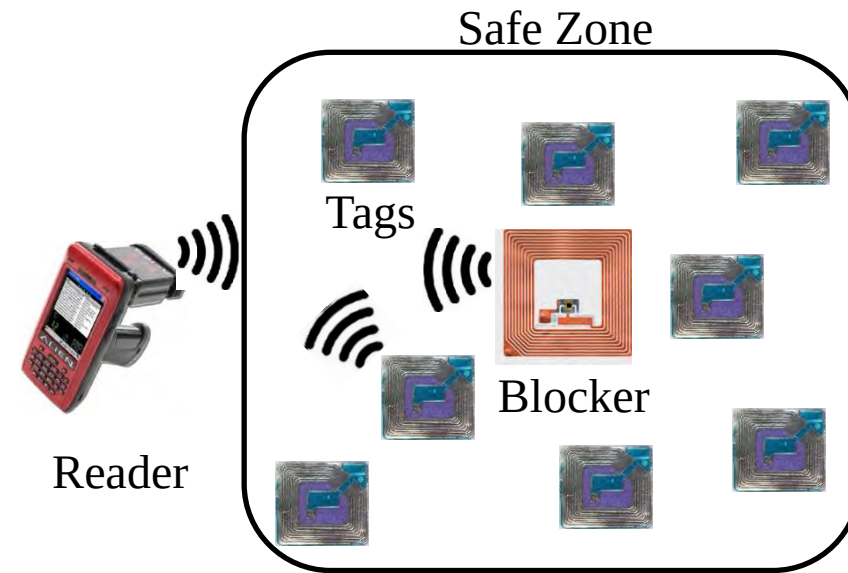
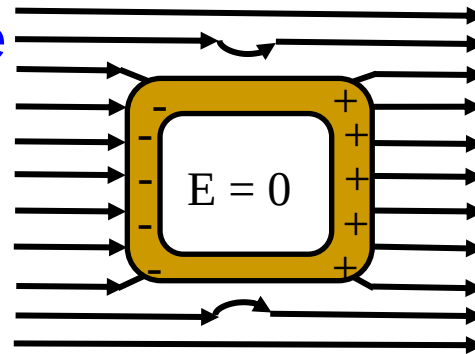
Source: S. Ghosh, J. Goswami, A. Majumder, A. Kumar, **S. P. Mohanty**, and B. K. Bhattacharyya, "Swing-Pay: One Card Meets All User Payment and Identity Needs", *IEEE Consumer Electronics Magazine (MCE)*, Volume 6, Issue 1, January 2017, pp. 82--93.

RFID Cybersecurity - Solutions

Selected RFID Security Methods



Faraday Cage



Blocker Tags

Source: Khattab 2017, Springer 2017 RFID Security

Drawbacks of Existing Cybersecurity Solutions



IT Cybersecurity Solutions Can't be Directly Extended to IoT/CPS Cybersecurity

IT Cybersecurity

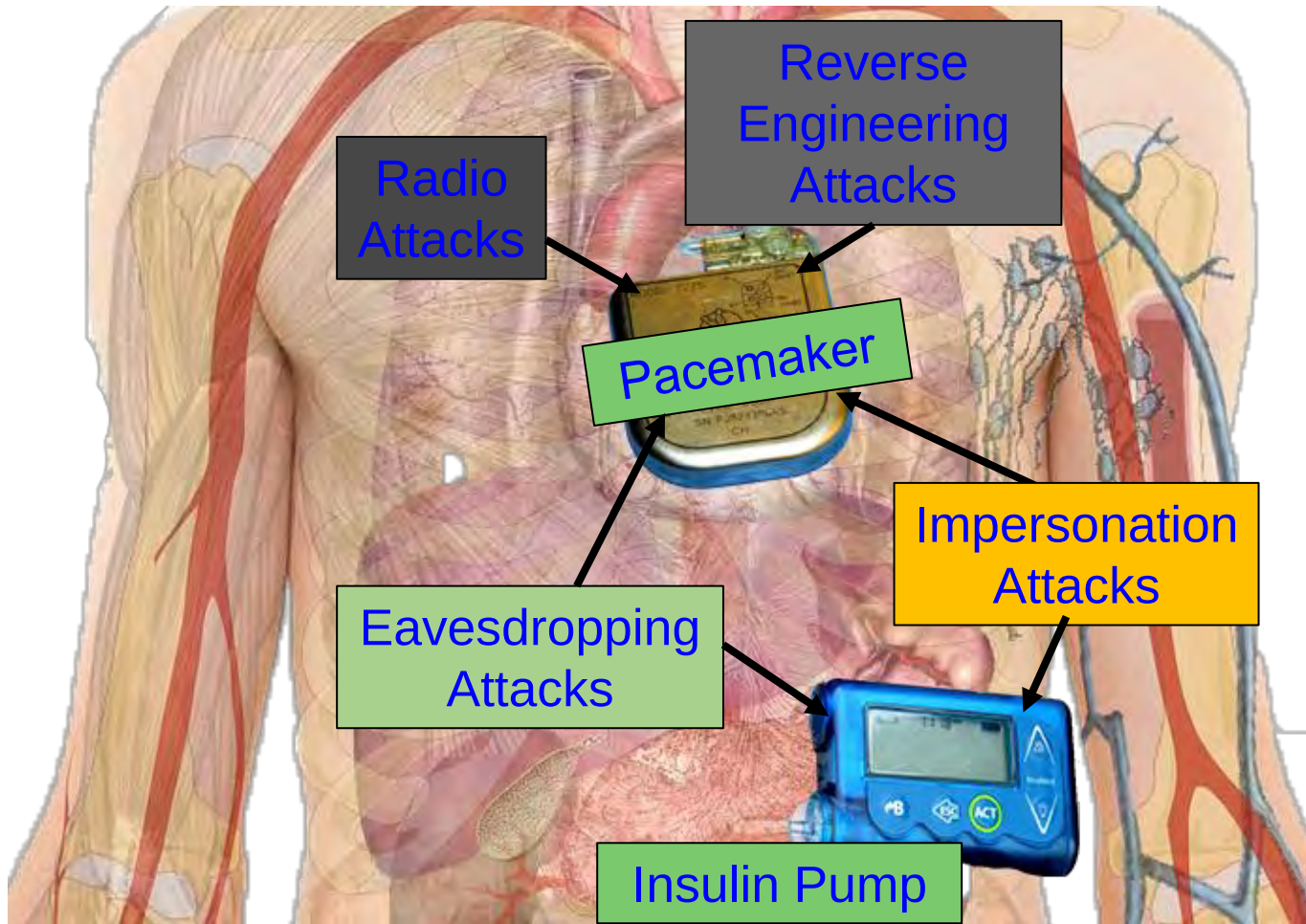
- IT infrastructure may be well protected rooms
- Limited variety of IT network devices
- Millions of IT devices
- Significant computational power to run heavy-duty security solutions
- IT security breach can be costly

IoT Cybersecurity

- IoT may be deployed in open hostile environments
- Significantly large variety of IoT devices
- Billions of IoT devices
- May not have computational power to run security solutions
- IoT security breach (e.g. in a IoMT device like pacemaker, insulin pump) can be life threatening

Maintaining of Cybersecurity of Electronic Systems, IoT, CPS, needs **Energy**, and affects performance.

Cybersecurity Measures in Healthcare Cyber-Physical Systems is Hard

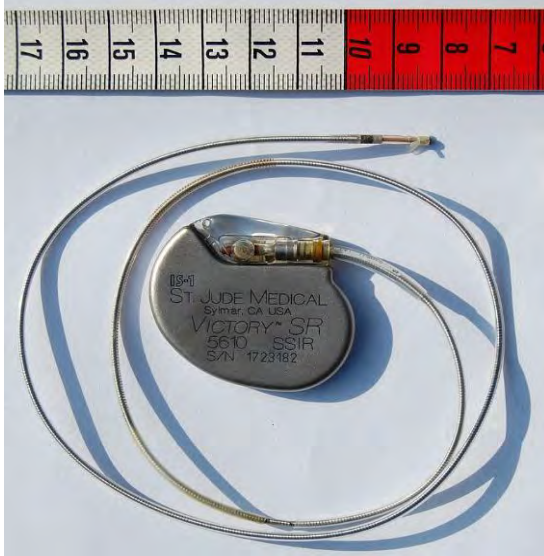


Collectively (WMD+IMD):
Implantable and Wearable
Medical Devices (IWMDs)

Implantable and Wearable Medical
Devices (IWMDs):

- Longer Battery life
- Safer device
- Smaller size
- Smaller weight
- Not much computational capability

H-CPS Cybersecurity Measures is Hard - Energy Constrained



Pacemaker
Battery Life
- 10 years



Neurostimulator
Battery Life
- 8 years

- Implantable Medical Devices (IMDs) have integrated battery to provide energy to all their functions → Limited Battery Life depending on functions
- Higher battery/energy usage → Lower IMD lifetime
- Battery/IMD replacement → Needs surgical risky procedures

Source: C. Camara, P. Peris-Lopeza, and J. E. Tapiadora, "Security and privacy issues in implantable medical devices: A comprehensive survey", *Elsevier Journal of Biomedical Informatics*, Volume 55, June 2015, Pages 272-289.

Cybersecurity Attacks – Software Vs Hardware Based

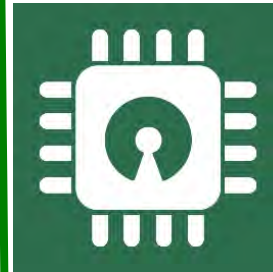
Software Based

- Software attacks via communication channels
- Typically from remote
- More frequent
- Selected Software based:
 - ❑ Denial-of-Service (DoS)
 - ❑ Routing Attacks
 - ❑ Malicious Injection
 - ❑ Injection of fraudulent packets
 - ❑ Snooping attack of memory
 - ❑ Spoofing attack of memory and IP address
 - ❑ Password-based attacks



Hardware Based

- Hardware or physical attacks
- Maybe local
- More difficult to prevent
- Selected Hardware based:
 - ❑ Hardware backdoors (e.g. Trojan)
 - ❑ Inducing faults
 - ❑ Electronic system tampering/ jailbreaking
 - ❑ Eavesdropping for protected memory
 - ❑ Side channel attack
 - ❑ Hardware counterfeiting



Source: Mohanty ICCE Panel 2018

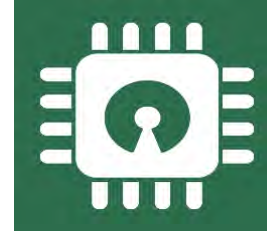
Cybersecurity Solutions – Software Vs Hardware Based

Software Based



- Introduces latency in operation
- Flexible - Easy to use, upgrade and update
- Wider-Use - Use for all devices in an organization
- Higher recurring operational cost
- Tasks of encryption easy compared to hardware – substitution tables
- Needs general purpose processor
- Can't stop hardware reverse engineering

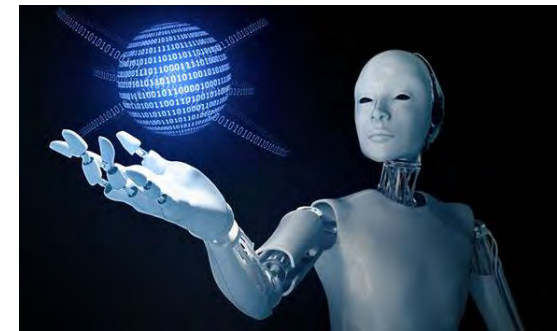
Source: Mohanty ICCE Panel 2018



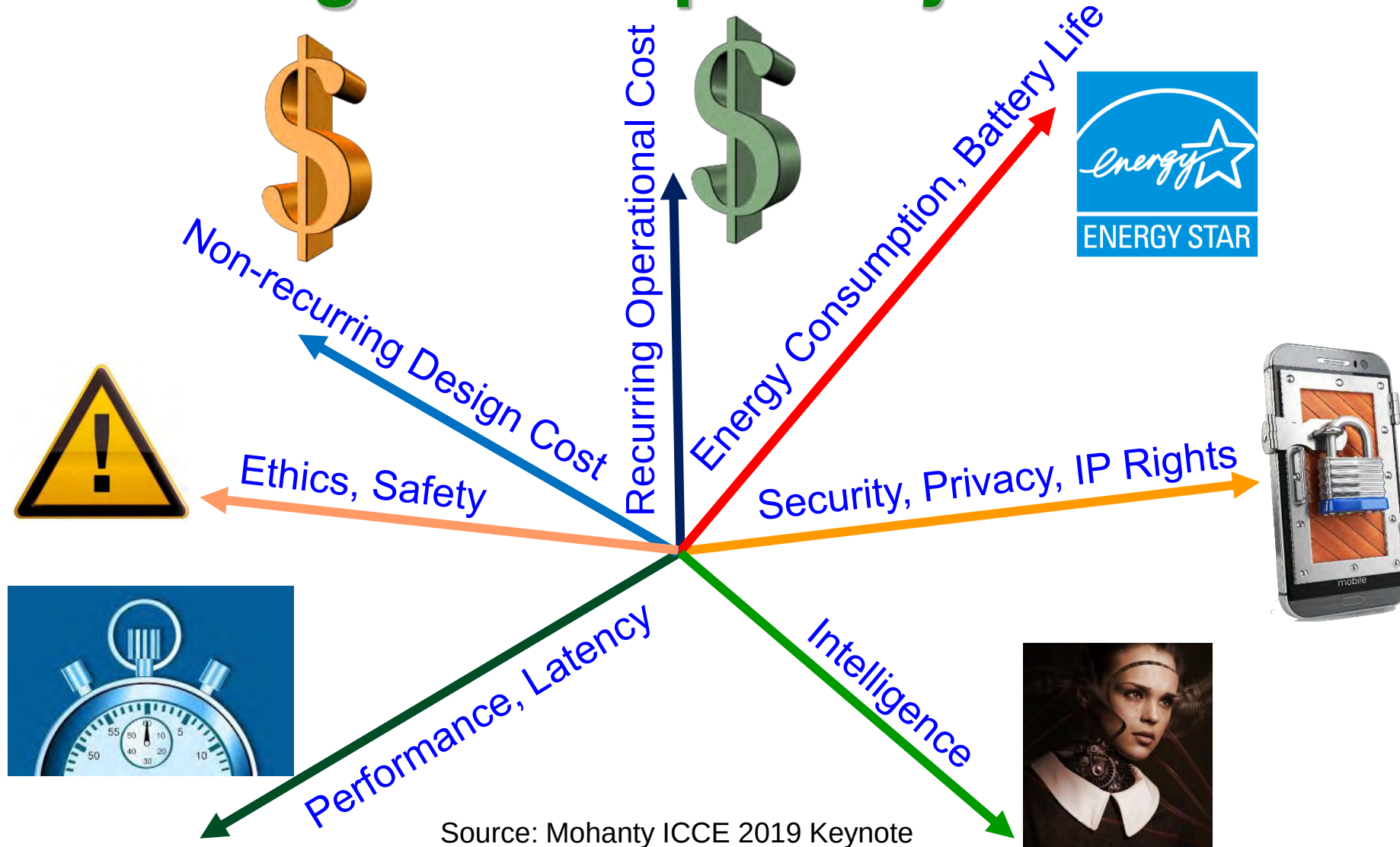
Hardware Based

- High-Speed operation
- Energy-Efficient operation
- Low-cost using ASIC and FPGA
- Tasks of encryption easy compared to software – bit permutation
- Easy integration in CE systems
- Possible security at source-end like sensors, better suitable for IoT
- Susceptible to side-channel attacks
- Can't stop software reverse engineering

Security-by-Design (SbD) – The Principle



CPS Design - Multiple Objectives for Sustainability



Source: Mohanty ICCE 2019 Keynote

Smart Cities
Vs
Smart Villages

Security by Design (SbD)

Embedding of security/privacy into the architecture (hardware+software) of various products, programs, or services.

Retrofitting: Difficult → Impossible!



Source: <https://teachprivacy.com/tag/privacy-by-design/>

Security by Design (SbD)



7 Fundamental Principles

Proactive not Reactive

Security/Privacy as the Default

Security/Privacy Embedded into Design

Full Functionality - Positive-Sum, not Zero-Sum

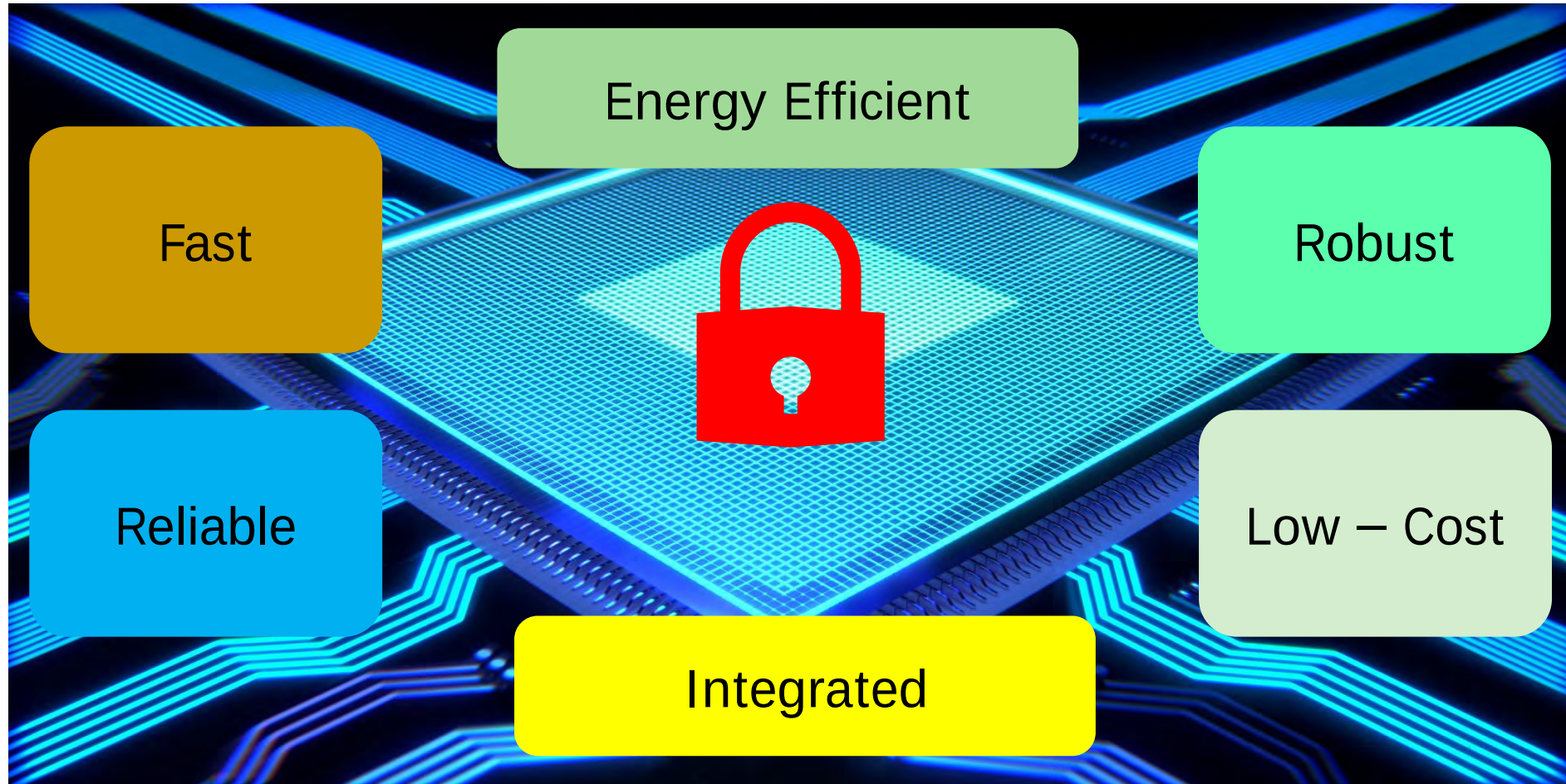
End-to-End Security/Privacy - Lifecycle Protection

Visibility and Transparency

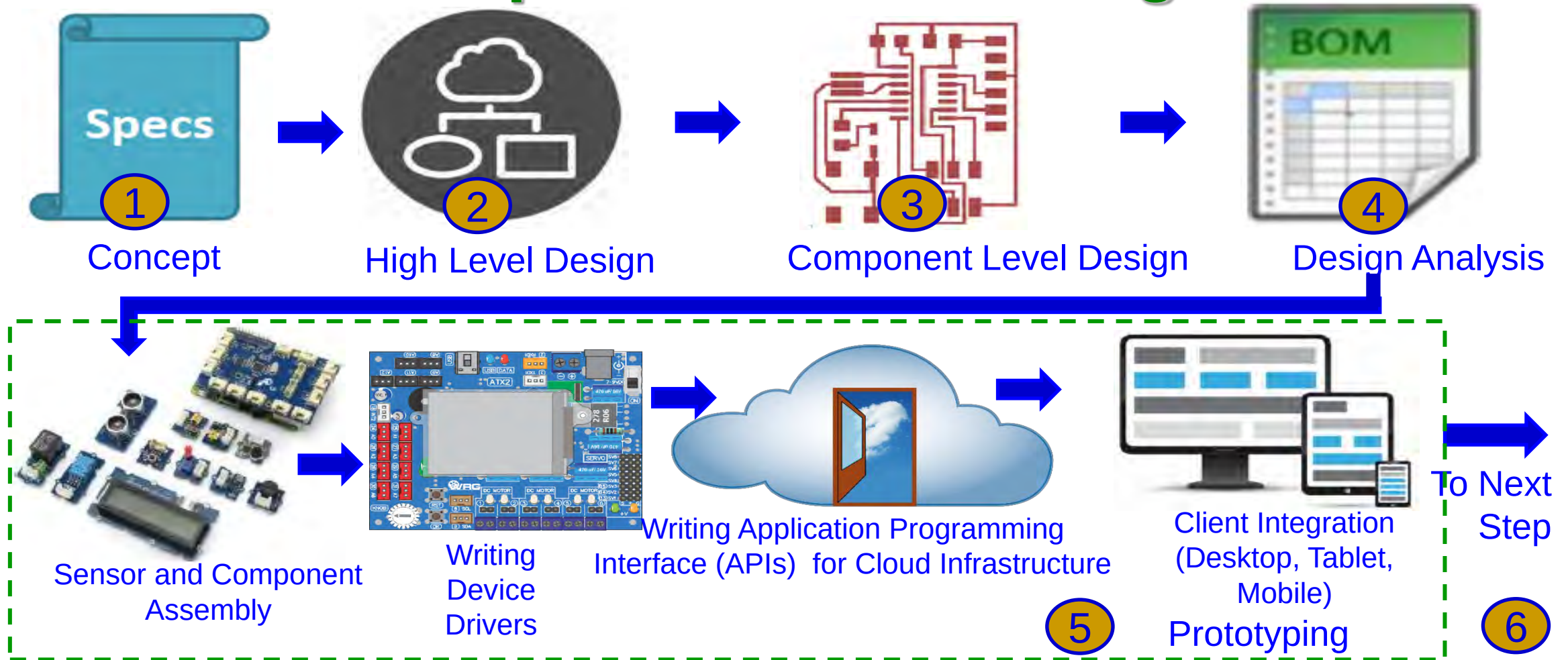
Respect for Users

Source: https://iapp.org/media/pdf/resource_center/Privacy%20by%20Design%20-%207%20Foundational%20Principles.pdf

Security-by-Design (SbD) or Hardware Assisted Security (HAS) - Advantages



SbD Principle – IoT/CPS Design Flow



How to integrate cybersecurity and privacy at every stage of design flow?

Source: <http://events.linuxfoundation.org/sites/events/files/slides/Design%20-%20End-to-End%20IoT%20Solution%20-%20Shivakumar%20Mathapathi.pdf>

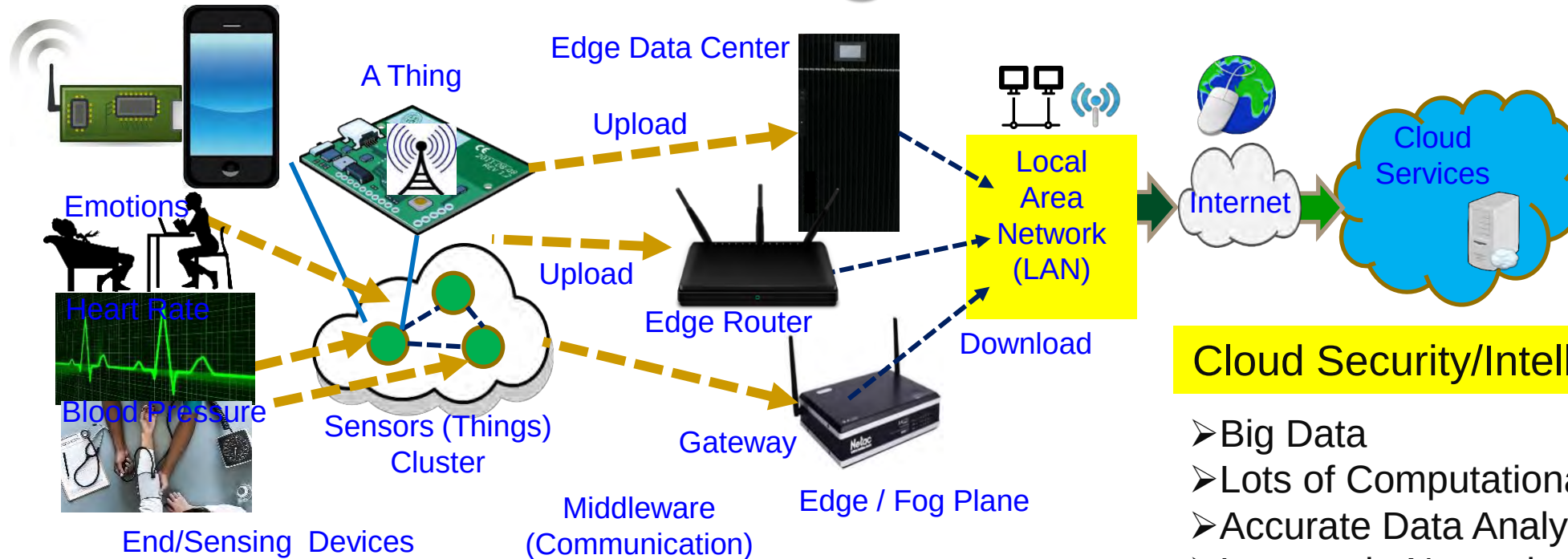
SbD Principle – IoT/CPS Design Flow



How to validate and document cybersecurity and privacy features at every stage of production?

Source: <http://events.linuxfoundation.org/sites/events/files/slides/Design%20-%20End-to-End%20IoT%20Solution%20-%20Shivakumar%20Mathapathi.pdf>

CPS – IoT-Edge Vs IoT-Cloud



End Security/Intelligence

- Minimal Data
- Minimal Computational Resource
- Least Accurate Data Analytics
- Very Rapid Response

Edge Security/Intelligence

- Less Data
- Less Computational Resource
- Less Accurate Data Analytics
- Rapid Response

Cloud Security/Intelligence

- Big Data
- Lots of Computational Resource
- Accurate Data Analytics
- Latency in Network
- Energy Overhead in Communications

Heavy-Duty ML is more suitable for smart cities

TinyML at End and/or Edge is key for smart villages.

Hardware Cybersecurity Primitives

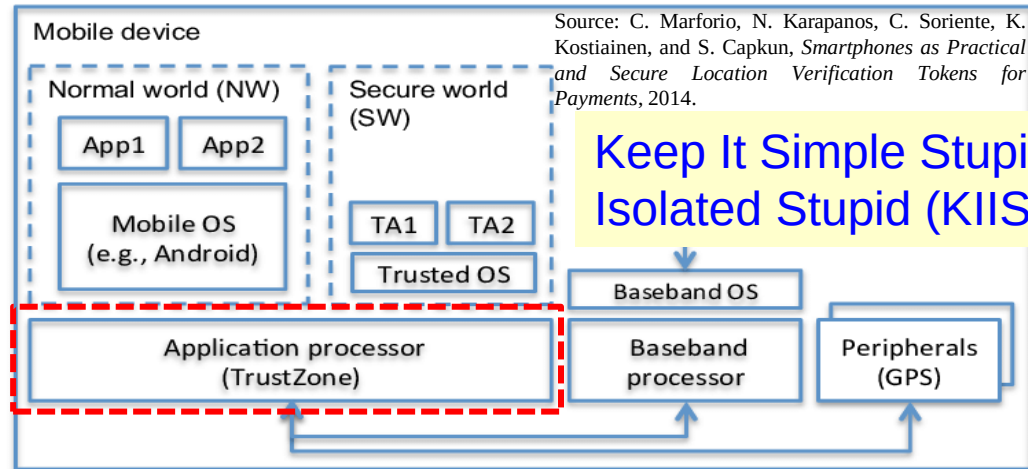
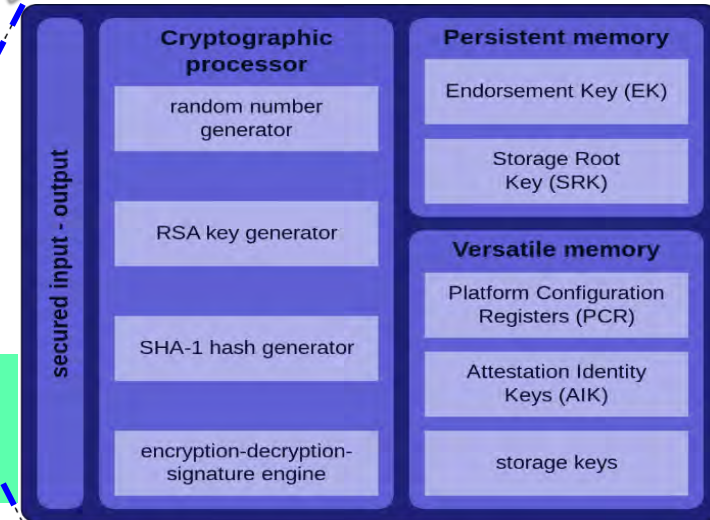
– HSM, TrustZone, TPM, and PUF



Hardware Security Module (HSM)



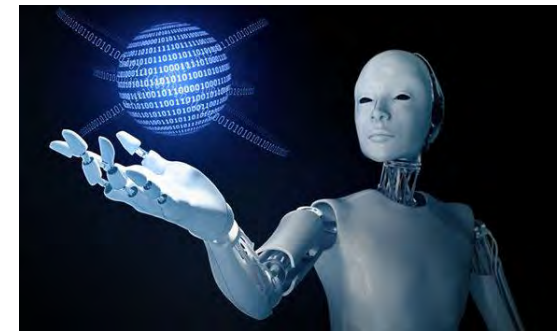
Trusted Platform Module (TPM)



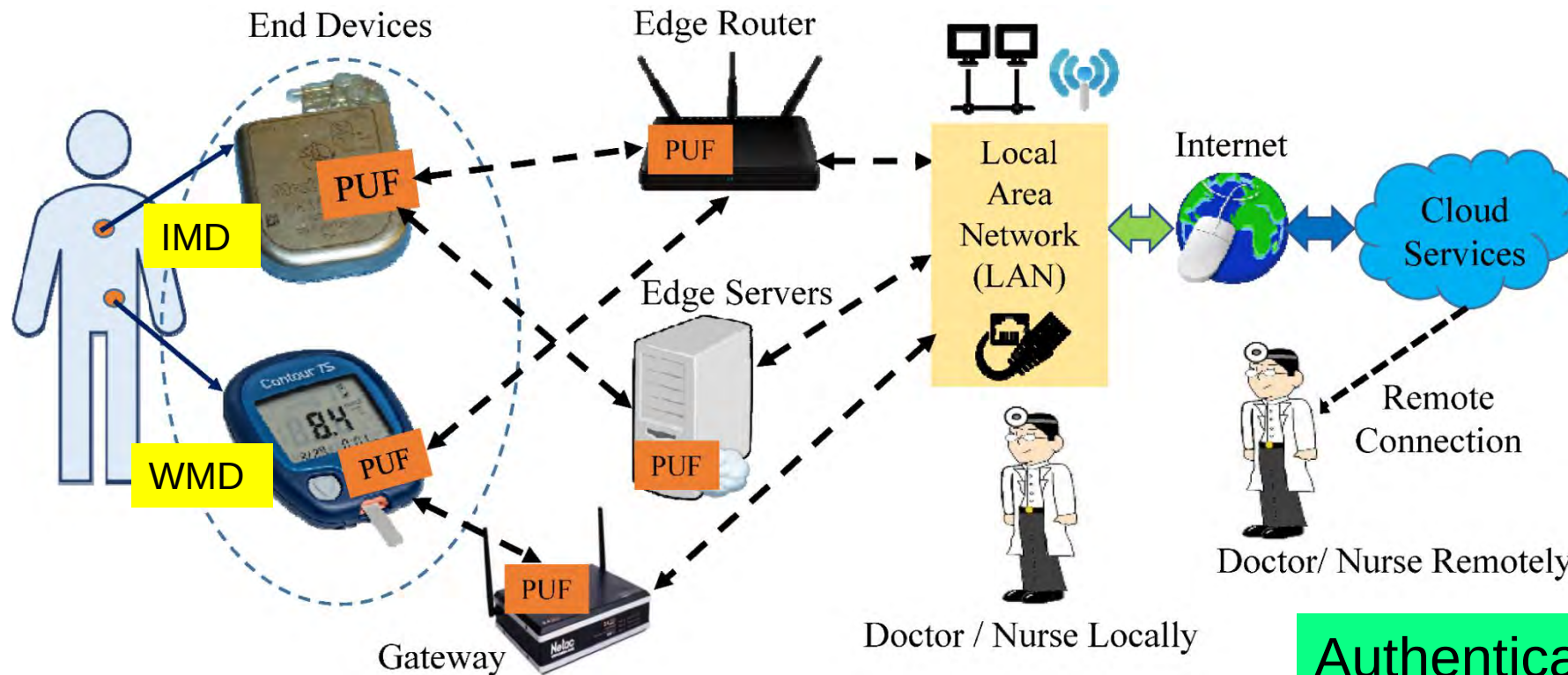
Physical Unclonable Functions (PUF)

Source: Electric Power Research Institute (EPRI)

Security-by-Design (SbD) – Specific Examples



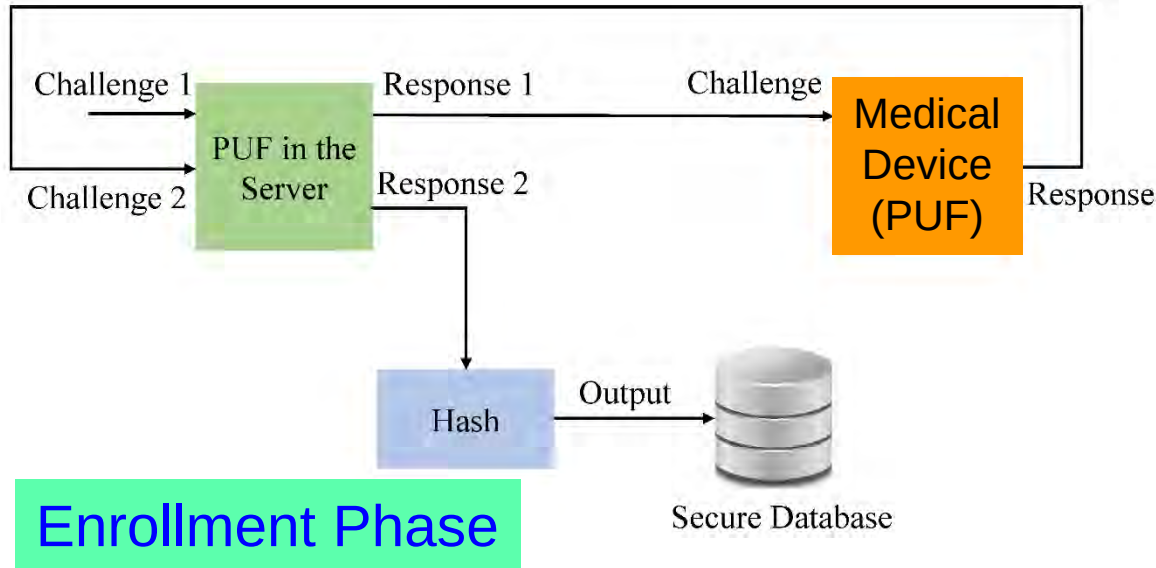
PMsec: Our Secure by Design Approach for Robust Security in Healthcare CPS



Authenticates Time - 1 sec
Power Consumption - 200 μ W

Source: V. P. Yanambaka, S. P. Mohanty, E. Kougianos, and D. Puthal, "PMsec: Physical Unclonable Function-Based Robust and Lightweight Authentication in the Internet of Medical Things", *IEEE Transactions on Consumer Electronics (TCE)*, Volume 65, Issue 3, August 2019, pp. 388--397.

IoMT Security – Our Proposed PMsec



PUF Security Full Proof:

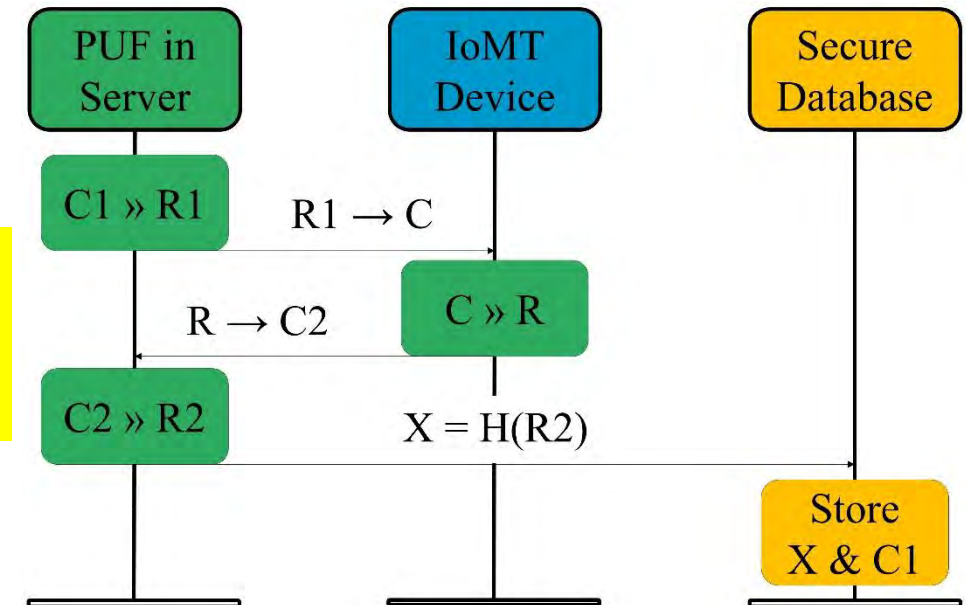
- Only server PUF Challenges are stored, not Responses
- Impossible to generate Responses without PUF

Source: V. P. Yanambaka, S. P. Mohanty, E. Kougianos, and D. Puthal, "PMsec: Physical Unclonable Function-Based Robust and Lightweight Authentication in the Internet of Medical Things", *IEEE Transactions on Consumer Electronics (TCE)*, Volume 65, Issue 3, August 2019, pp. 388--397.

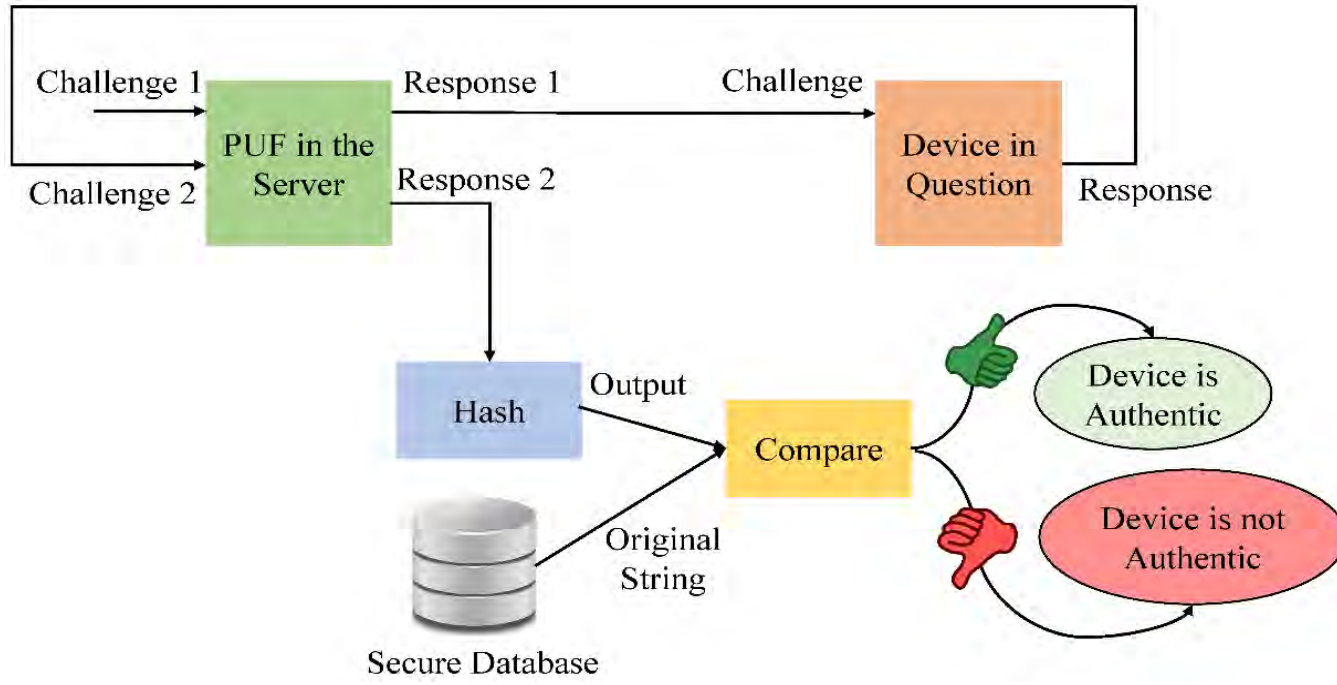
At the Doctor

- When a new IoMT-Device comes for an User

Device Registration Procedure



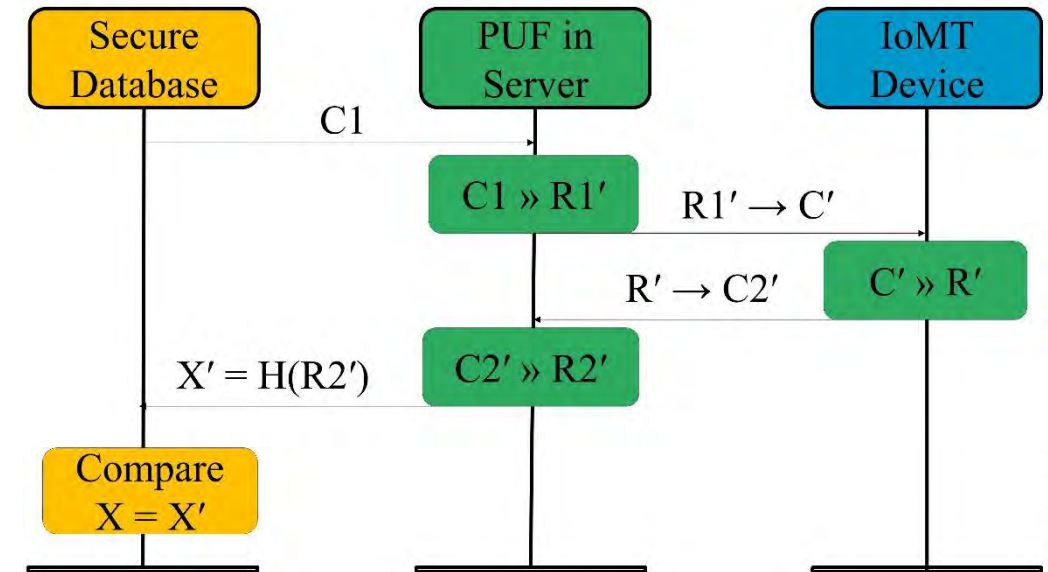
IoMT Security – Our Proposed PMsec



Authentication Phase

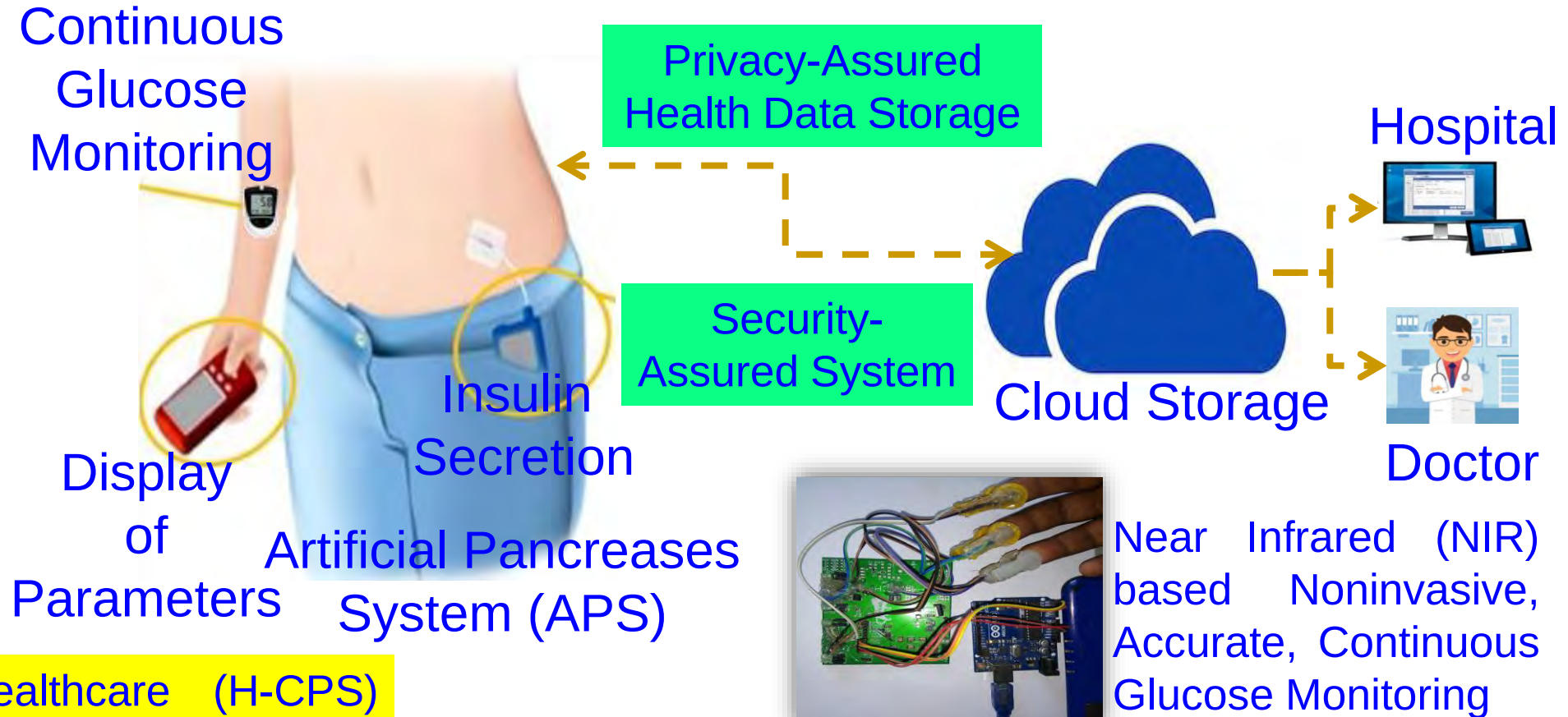
At the Doctor
➤ When doctor needs to access an existing IoMT-device

Device Authentication Procedure



Source: V. P. Yanambaka, S. P. Mohanty, E. Kougianos, and D. Puthal, "PMsec: Physical Unclonable Function-Based Robust and Lightweight Authentication in the Internet of Medical Things", *IEEE Transactions on Consumer Electronics (TCE)*, Volume 65, Issue 3, August 2019, pp. 388--397.

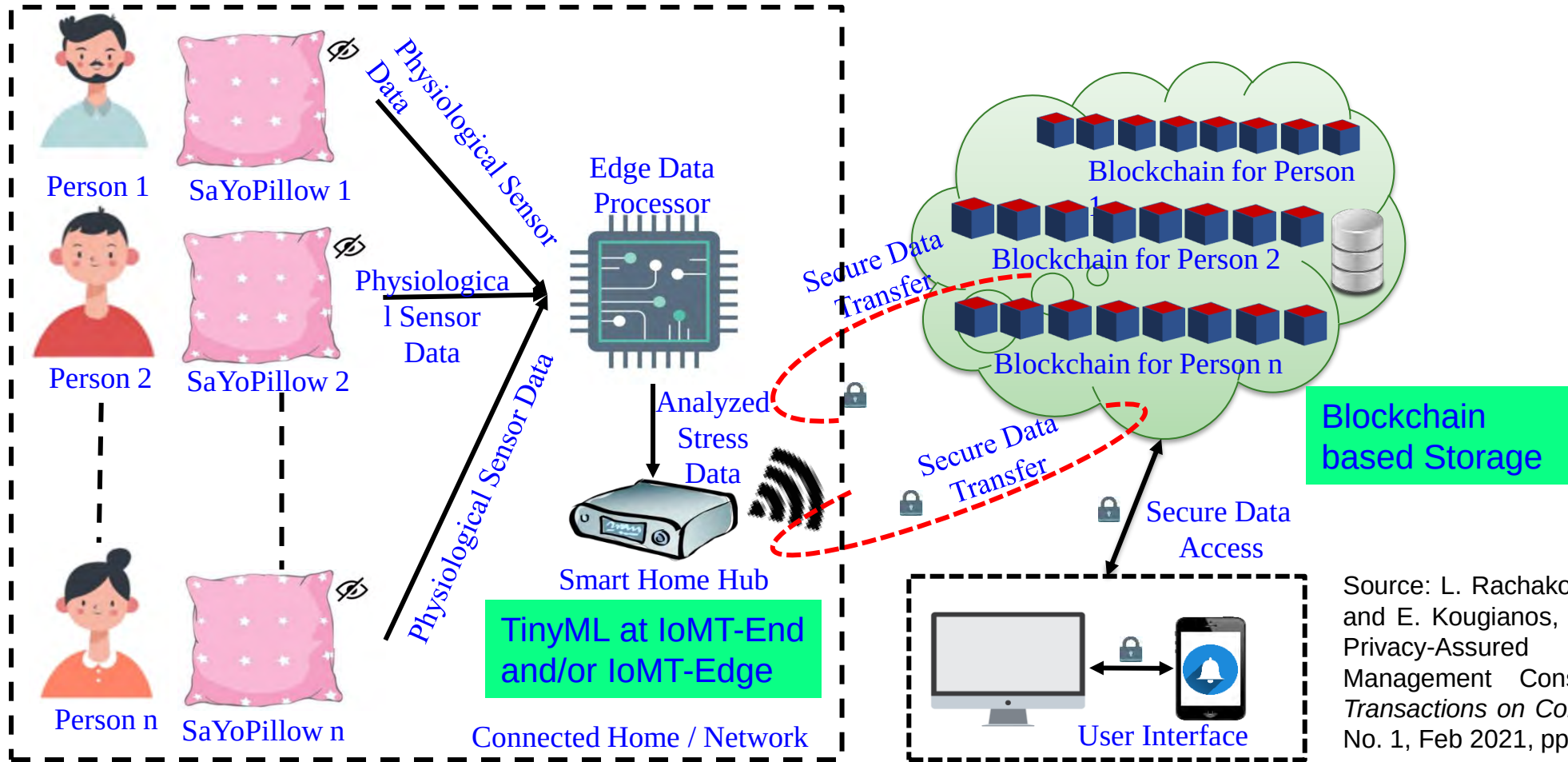
Secure-iGLU - Our Intelligent Non-Invasive Glucose Monitoring with Insulin Control Device



Smart Healthcare (H-CPS)
→ Security, Privacy, ...

P. Jain, A. M. Joshi, and S. P. Mohanty, "iGLU: An Intelligent Device for Accurate Non-Invasive Blood Glucose-Level Monitoring in Smart Healthcare", *IEEE Consumer Electronics Magazine (MCE)*, Vol. 9, No. 1, January 2020, pp. 35–42.

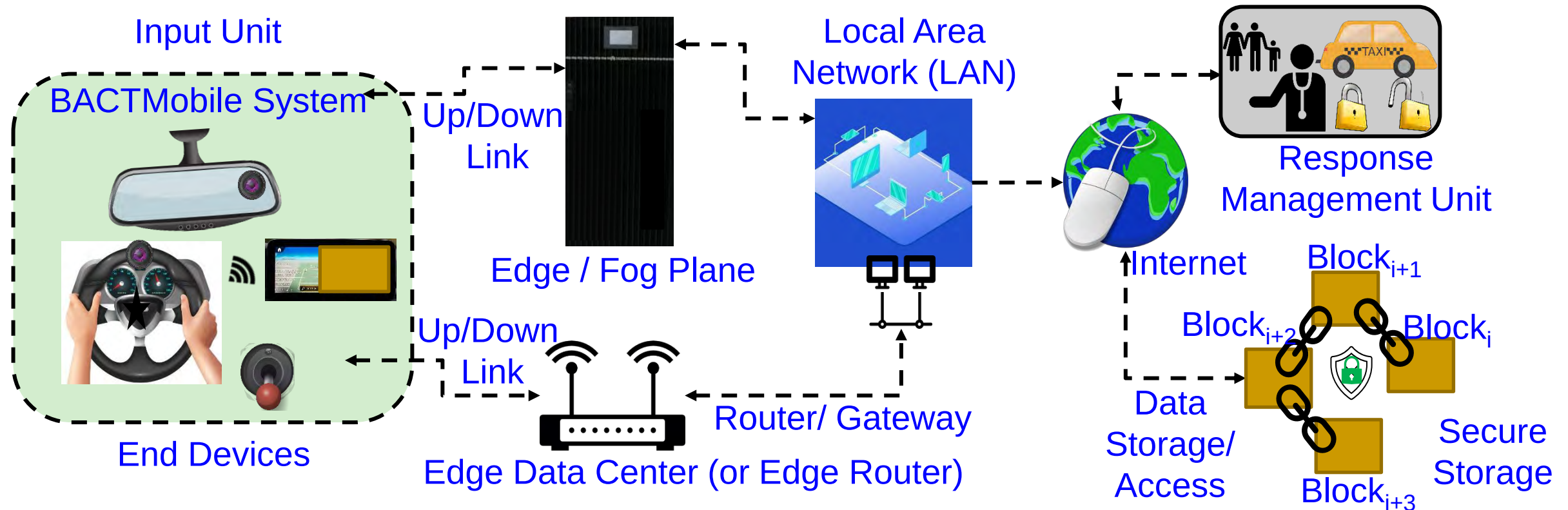
Our Smart-Yoga Pillow (SaYoPillow) with TinyML and Blockchain based Security



Source: L. Rachakonda, A. K. Bapatla, S. P. Mohanty, and E. Kougianos, "SaYoPillow: Blockchain-Integrated Privacy-Assured IoMT Framework for Stress Management Considering Sleeping Habit", *IEEE Transactions on Consumer Electronics (TCE)*, Vol. 67, No. 1, Feb 2021, pp. 20-29.

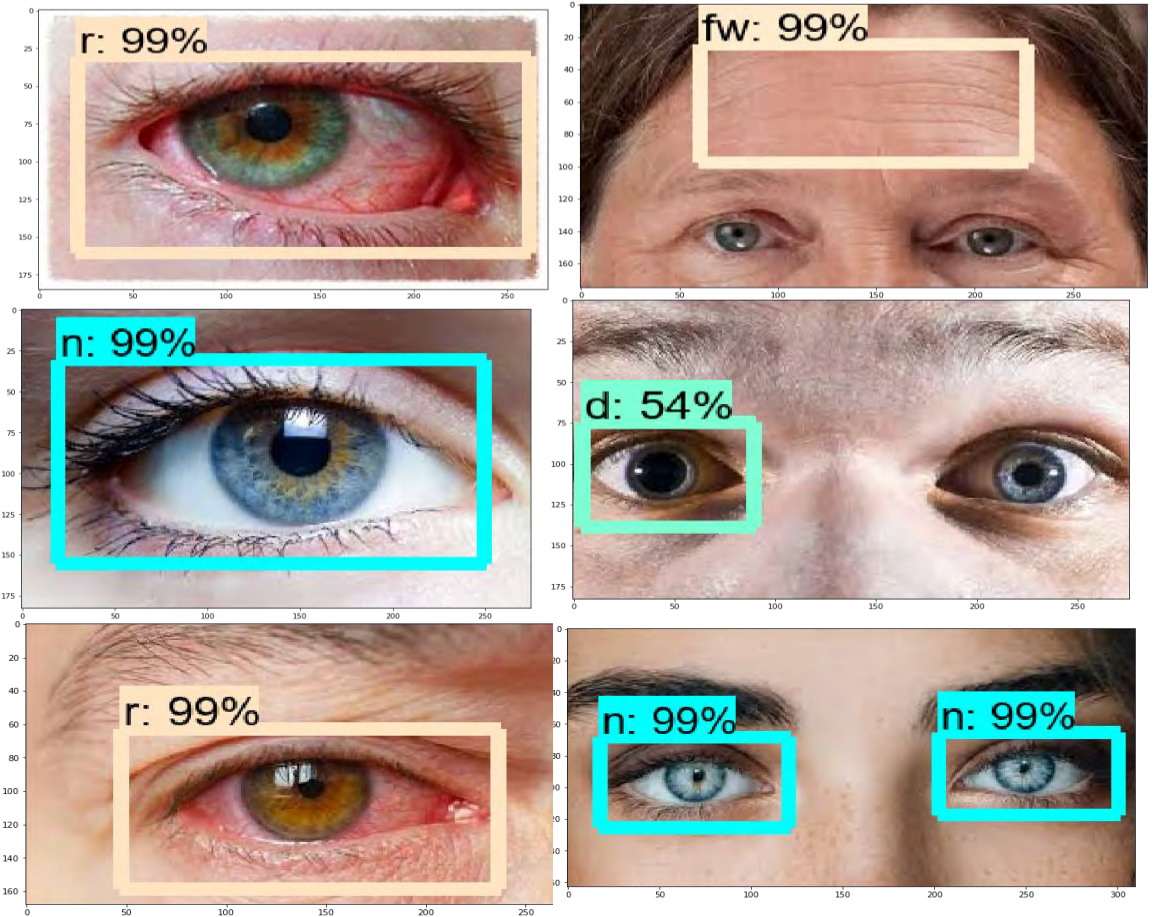
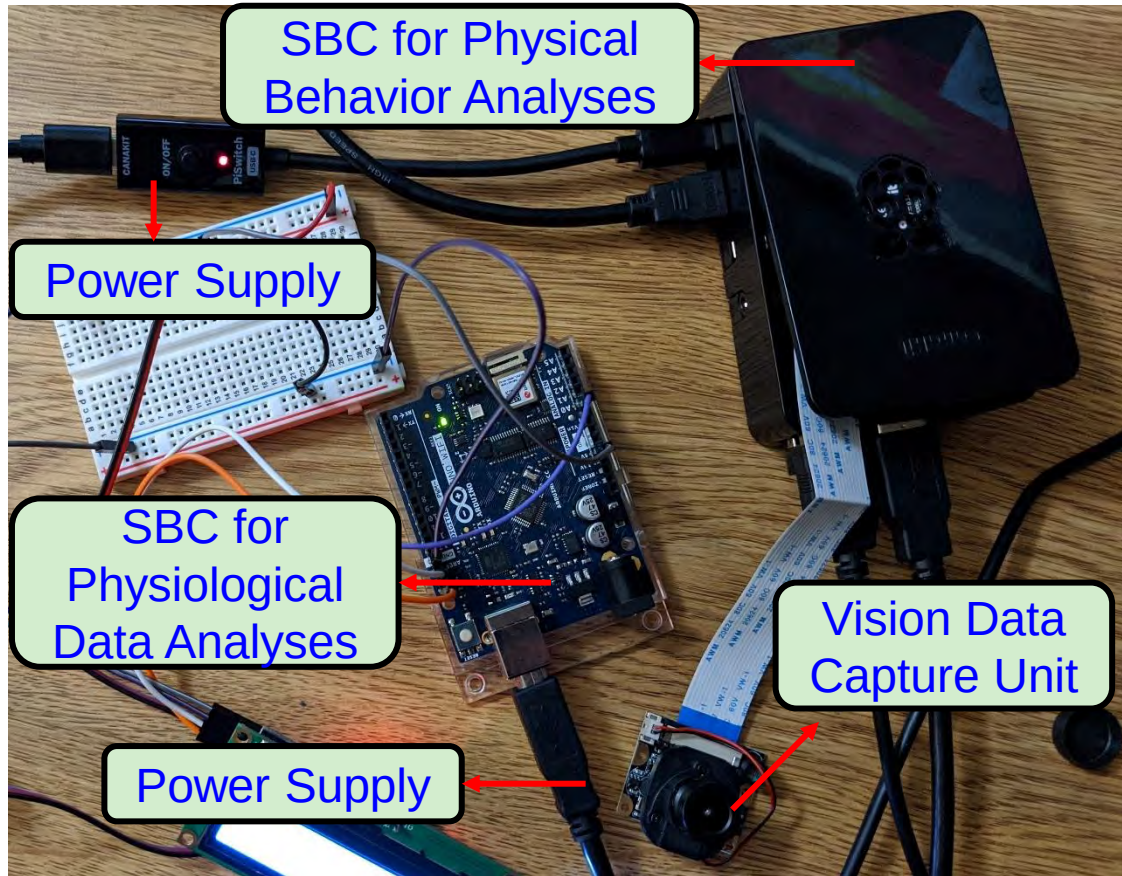
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Our Smart Blood Alcohol Concentration Tracking Mechanism in Healthcare CPS - BACTmobile



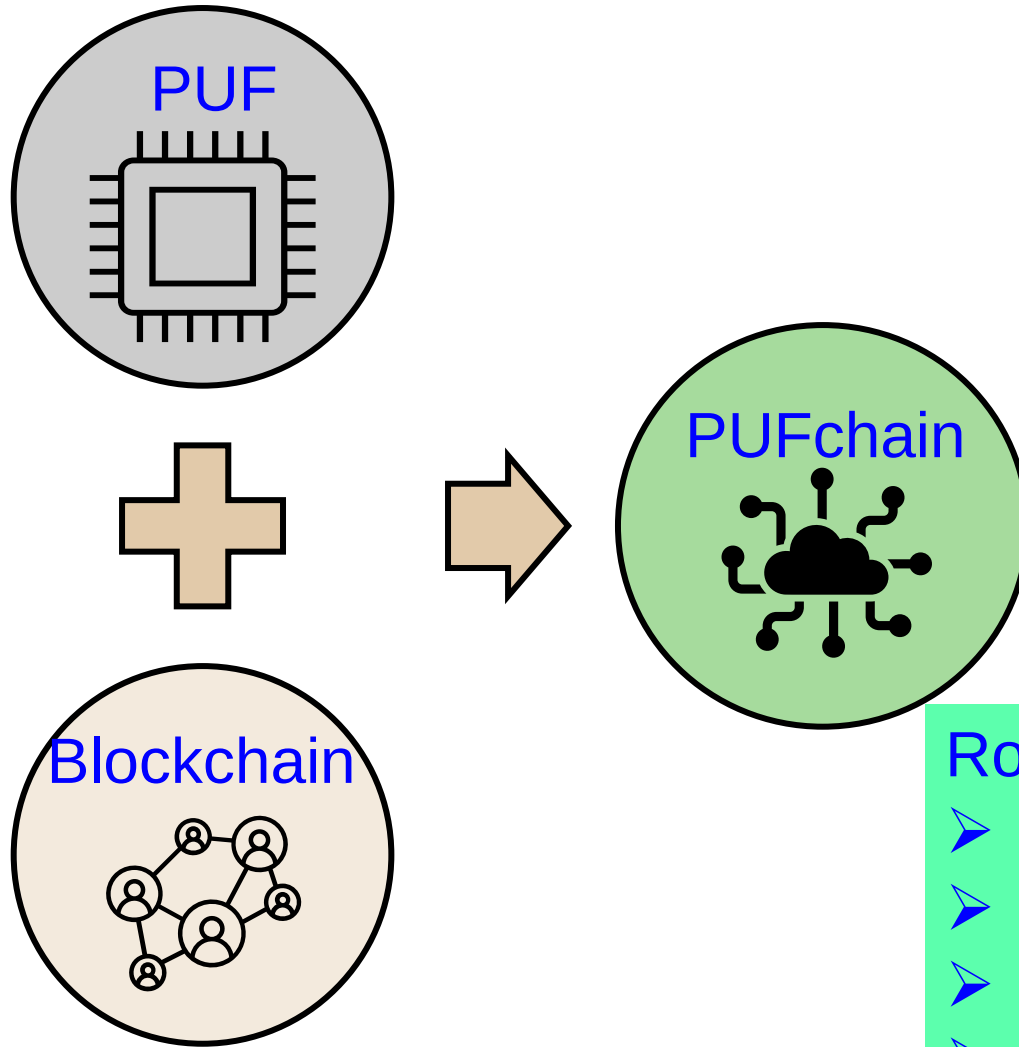
Source: L. Rachakonda, A. K. Bapatla, **S. P. Mohanty**, and E. Kougianos, "BACTmobile: A Smart Blood Alcohol Concentration Tracking Mechanism for Smart Vehicles in Healthcare CPS Framework", *Springer Nature Computer Science (SN-CS)*, Vol. 3, No. 3, May 2022, Article: 236, 24-pages, DOI: <https://doi.org/10.1007/s42979-022-01142-9>.

Our Smart Blood Alcohol Concentration Tracking Mechanism in Healthcare CPS - BACTmobile



Source: L. Rachakonda, A. K. Bapatla, **S. P. Mohanty**, and E. Kougianos, "BACTmobile: A Smart Blood Alcohol Concentration Tracking Mechanism for Smart Vehicles in Healthcare CPS Framework", *Springer Nature Computer Science (SN-CS)*, Vol. 3, No. 3, May 2022, Article: 236, 24-pages, DOI: <https://doi.org/10.1007/s42979-022-01142-9>.

PUFchain – The Big Idea

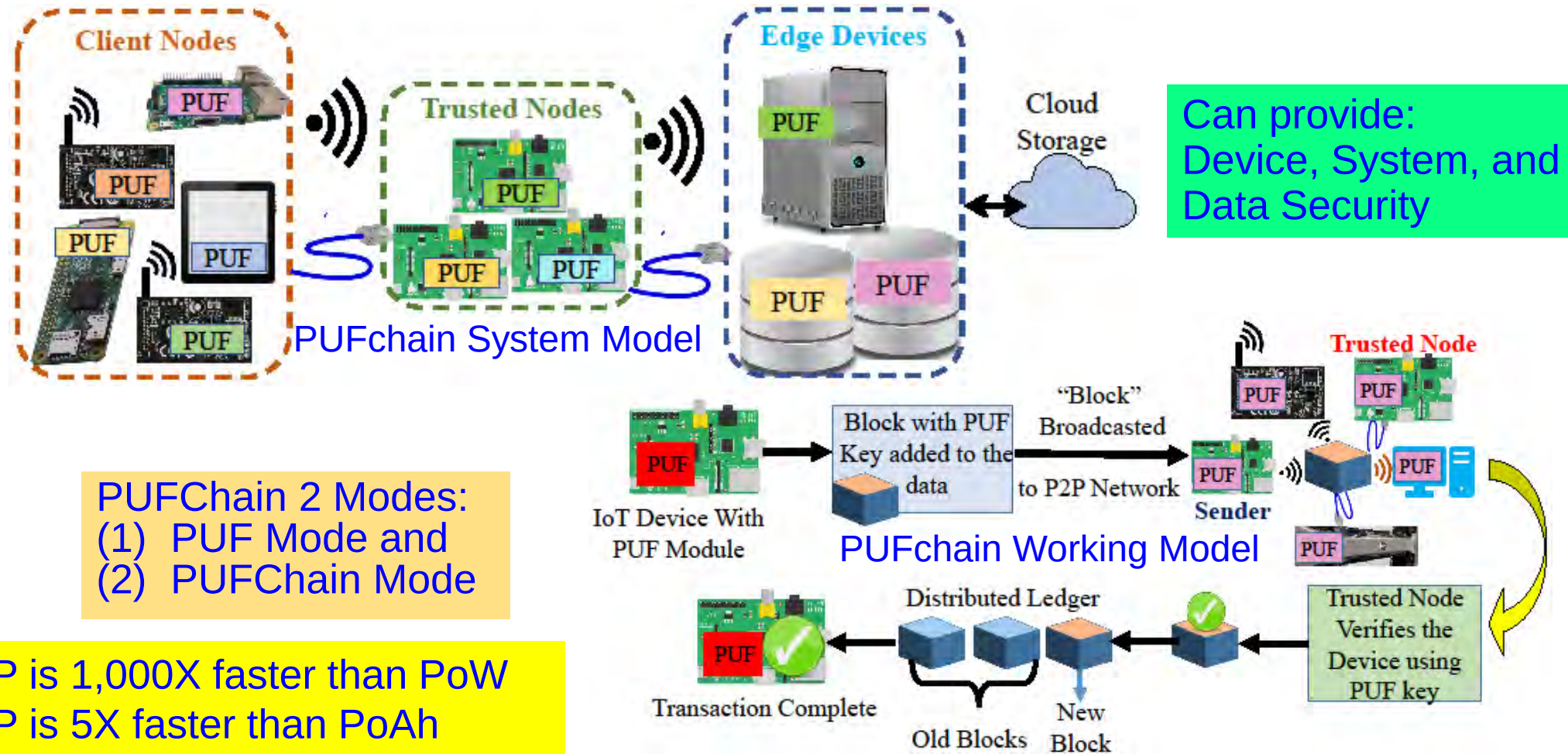


Blockchain Technology is integrated with Physically Unclonable Functions as PUFchain by storing the PUF Key into immutable Blockchain

Roles of PUF:

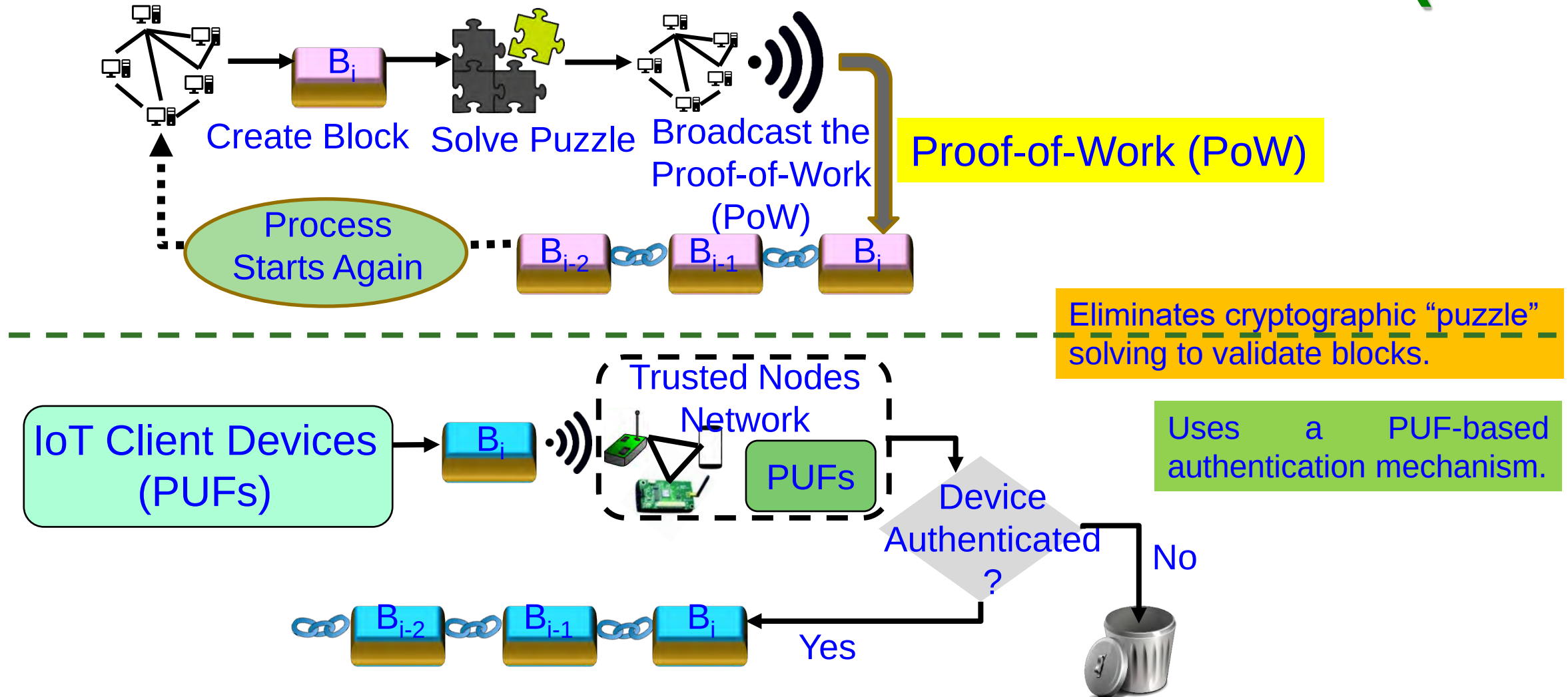
- Hardware Accelerator for Blockchain
- Independent Authentication
- Double-Layer Protection
- 3 modes: PUF, Blockchain, PUF+Blockchain

PUFchain: Our Hardware-Assisted Scalable Blockchain



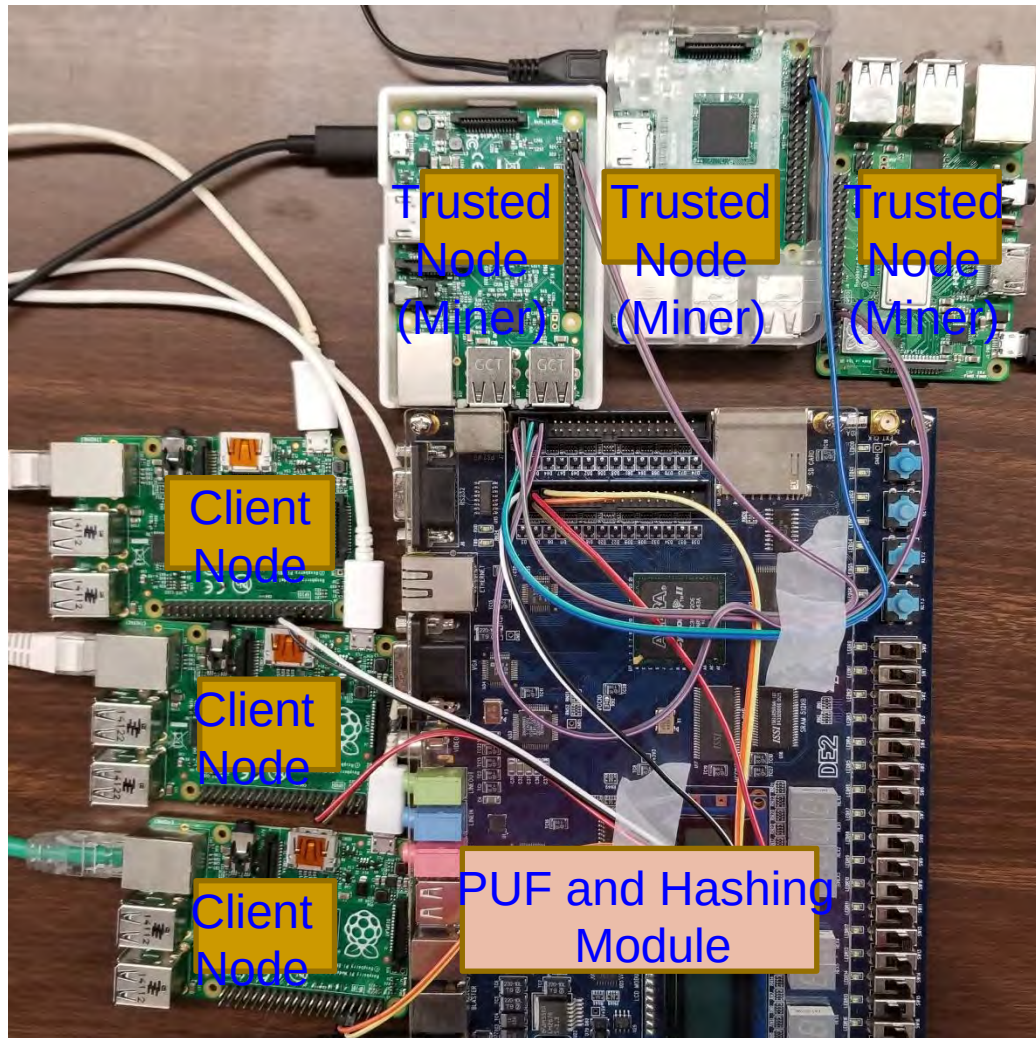
Source: S. P. Mohanty, V. P. Yanambaka, E. Kougianos, and D. Puthal, "PUFchain: Hardware-Assisted Blockchain for Sustainable Simultaneous Device and Data Security in Internet of Everything (IoE)", *IEEE Consumer Electronics Magazine (MCE)*, Vol. 9, No. 2, March 2020, pp. 8-16.

Our Proof-of-PUF-Enabled-Authentication (PoP)



Source: S. P. Mohanty, V. P. Yanambaka, E. Kougianos, and D. Puthal, "PUFchain: Hardware-Assisted Blockchain for Sustainable Simultaneous Device and Data Security in Internet of Everything (IoE)", *IEEE Consumer Electronics Magazine (MCE)*, Vol. 9, No. 2, March 2020, pp. 8-16.

PUFchain: Our PoP is 1000X Faster than PoW



PoW - 10
min in cloud

PoAh – 950ms
in Raspberry Pi

PoP - 192ms in
Raspberry Pi

High Power

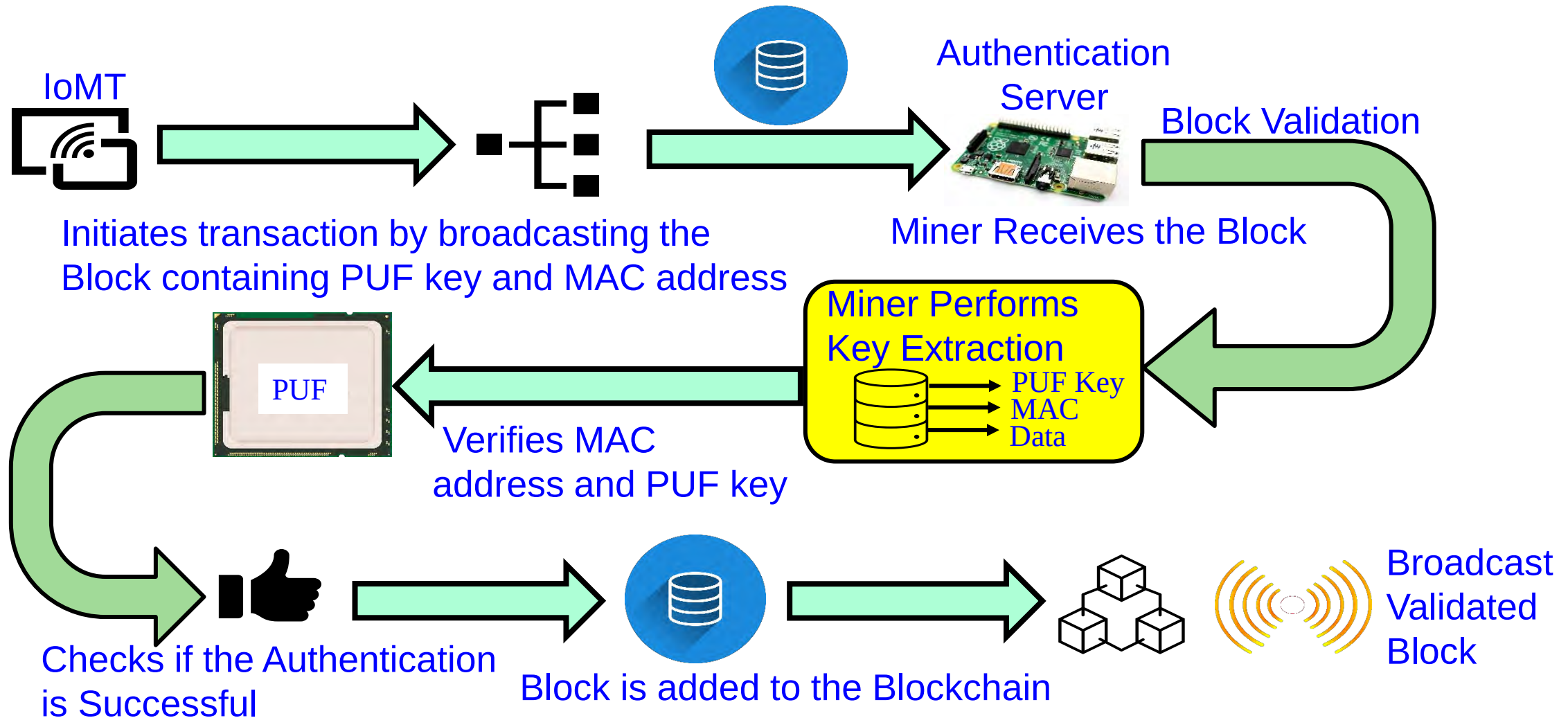
3 W Power

5 W Power

- ✓ PoP is 1,000X faster than PoW
- ✓ PoP is 5X faster than PoAh

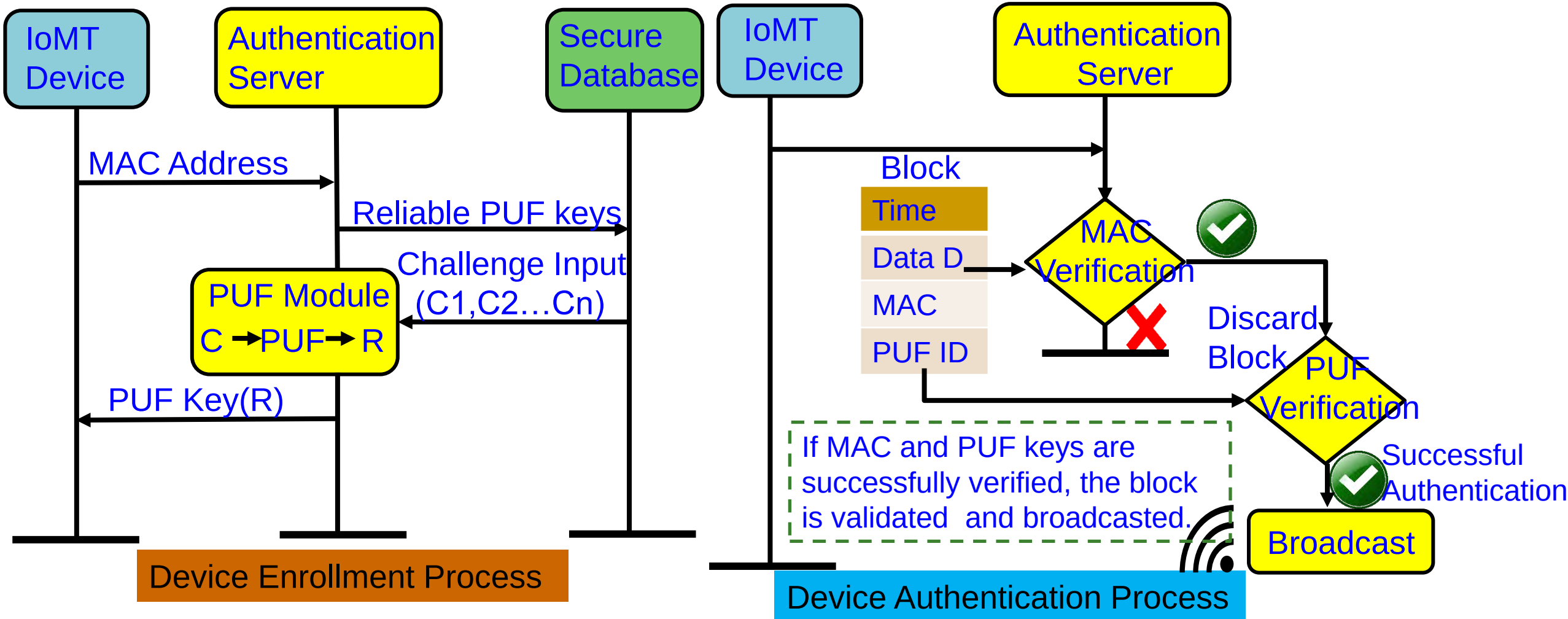
Source: S. P. Mohanty, V. P. Yanambaka, E. Kougianos, and D. Puthal, "PUFchain: Hardware-Assisted Blockchain for Sustainable Simultaneous Device and Data Security in Internet of Everything (IoE)", *IEEE Consumer Electronics Magazine (MCE)*, Vol. 9, No. 2, March 2020, pp. 8-16.

PUFchain 2.0: Our Hardware-Assisted Scalable Blockchain



Source: V. K. V. V. Bathalapalli, **S. P. Mohanty**, E. Kougianos, B. K. Baniya, and B. Rout, "PUFchain 2.0: Hardware-Assisted Robust Blockchain for Sustainable Simultaneous Device and Data Security in Smart Healthcare", *Springer Nature Computer Science (SN-CS)*, Vol. 3, No. 5, Sep 2022, Article: 344, 19-pages, DOI: <https://doi.org/10.1007/s42979-022-01238-2>.

PUFchain 2.0: PUF Integrated Blockchain ...



Source: V. K. V. V. Bathalapalli, **S. P. Mohanty**, E. Kougianos, B. K. Baniya, and B. Rout, "PUFchain 2.0: Hardware-Assisted Robust Blockchain for Sustainable Simultaneous Device and Data Security in Smart Healthcare", *Springer Nature Computer Science (SN-CS)*, Vol. 3, No. 5, Sep 2022, Article: 344, 19-pages, DOI: <https://doi.org/10.1007/s42979-022-01238-2>.

PUFchain 3.0: Comparative Analysis

Research Works	Application	DLT or Blockchain	Authentication Mechanism	Performance Metrics
Mohanty et al. 2020 - PUFchain	IoMT (Device and Data)	Blockchain	Proof-of-PUF-Enabled Authentication	PUF Design Uniqueness - 47.02%, Reliability-1.25%
Chaudhary et al. 2021 - Auto-PUFchain	Hawrdware Supply Chain	Blockchain	Smart Contracts	Gas Cost for Ethereum transaction 21.56 USD (5-Stage)
Al-Joboury et al. 2021 - PoQDB	IoT (Data)	Blockchain & Cobweb	IoT M2M Messaging (MQTT)	Transaction Time - 15 ms
Wang et al. 2022 - PUF-Based Authentication	IoMT (Device)	Blockchain	Smart Contracts	NA
Hellani et al. 2021- Tangle the Blockchain	IoT (Data)	Blockchain & Tangle	Smart Contracts	NA
Bathalapalli et al. 2022-PUFchain 2.0	IoMT (Device)	Blockchain	Media Access Control (MAC) & PUF based Authentication	Total On-Chip Power - 0.081 W, PUF Hamming Distance - 48.02 %
Our PUFchain 3.0 in 2022	IoMT (Device)	Tangle	Masked Authentication Messaging	Authentication 2.72 sec, Reliability - 100% (Approx), MAM Mode-Restricted

Source: V. K. V. V. Bathalapalli, **S. P. Mohanty**, E. Kougianos, B. K. Baniya, and B. Rout, “**PUFchain 3.0: Hardware-Assisted Distributed Ledger for Robust Authentication in the Internet of Medical Things**”, in *Proceedings of IFIP International Internet of Things Conference (IFIP-IoT)*, 2022, pp. 23--40, DOI: https://doi.org/10.1007/978-3-031-18872-5_2.

Smart Healthcare – Trustworthy Pharmaceutical Supply Chain

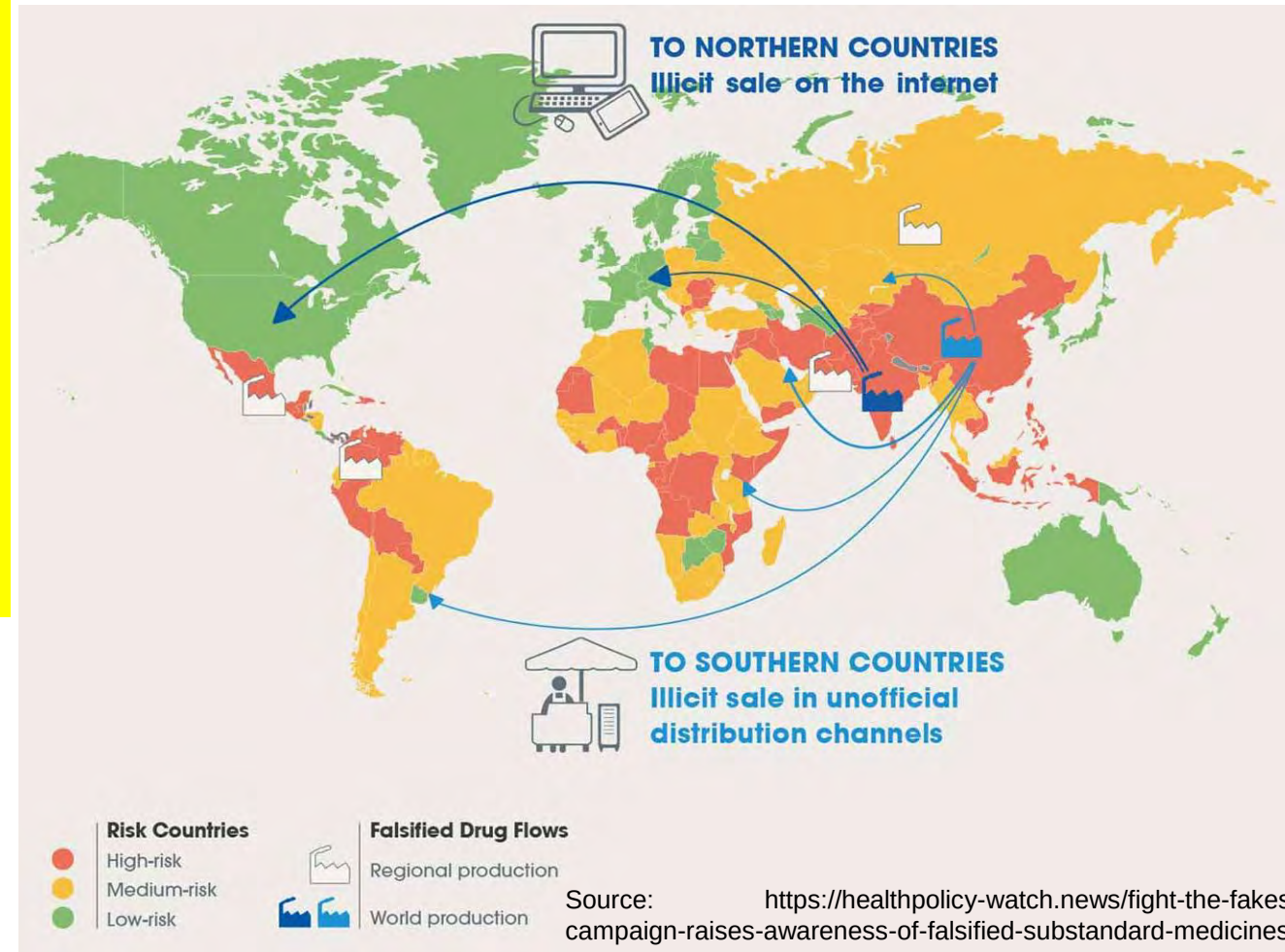
Fake Medicine - Serious Global Issue

- It is estimated that close to \$83 billion worth of counterfeit drugs are sold annually.
- One in 10 medical products circulating in developing countries are substandard or fake.
- In Africa: Counterfeit antimalarial drugs results in more than 120,000 deaths each year.
- USA has a closed drug distribution system intended to prevent counterfeits from entering U.S. markets, but it isn't foolproof due to many reason including illegal online pharmacy.

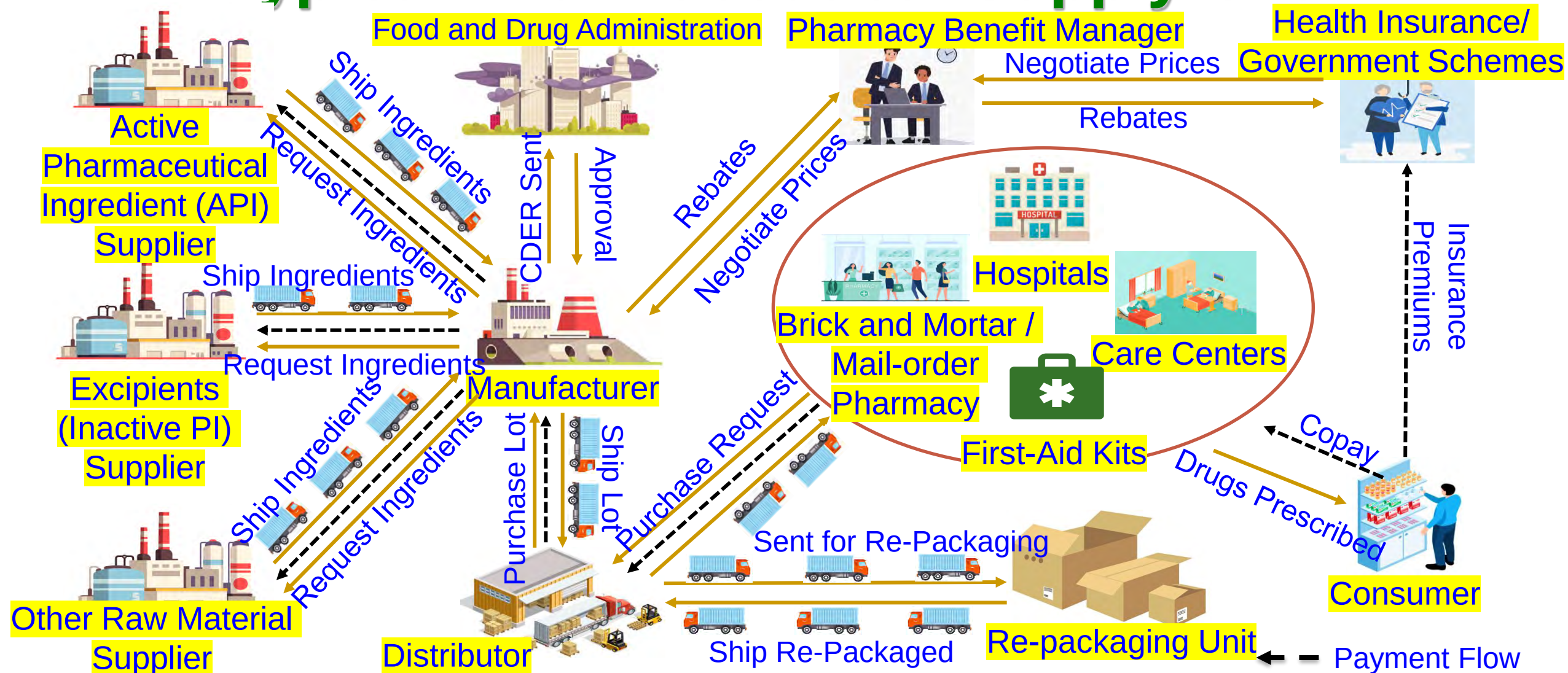
Source: <https://fraud.org/fakerx/fake-drugs-and-their-risks/counterfeit-drugs-are-a-global-problem/>



Source: <https://allaboutpharmacovigilance.org/be-aware-of-counterfeit-medicine/>



Typical Pharmaceutical Supply Chain



Source: Bapatla, A.K., et al.: PharmaChain: a blockchain to ensure counterfeit-free pharmaceutical supply chain. IET Netw. 1– 24 (2022). <https://doi.org/10.1049/ntw2.12041>

PharmaChain - Counterfeit Free Pharmaceutical

Enterprise Resource Planning

Transaction Ledger



Manufacturer places order and ingredients are supplied



Wholesaler places order from Manufacturer



Transfer of drugs from wholesaler to pharmacy



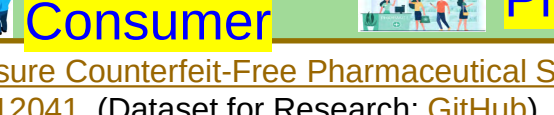
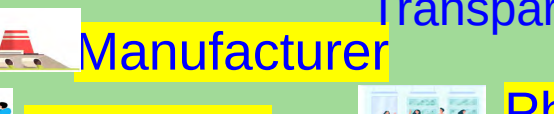
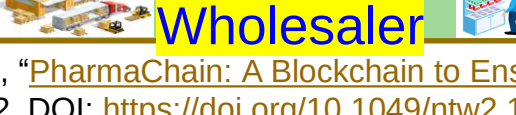
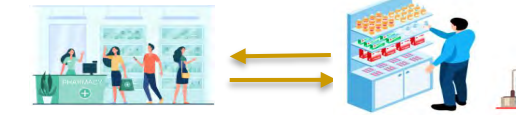
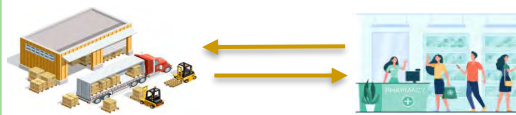
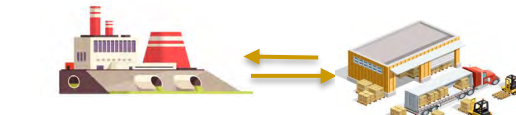
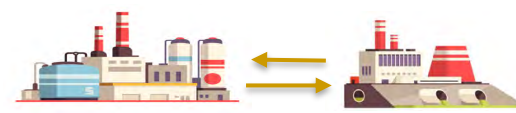
Prescribed medicines are dispensed to the consumer

Blind Parties



Blockchain System

Blockchain Ledger



Transparent Ledger

Ingredients

Manufacturer

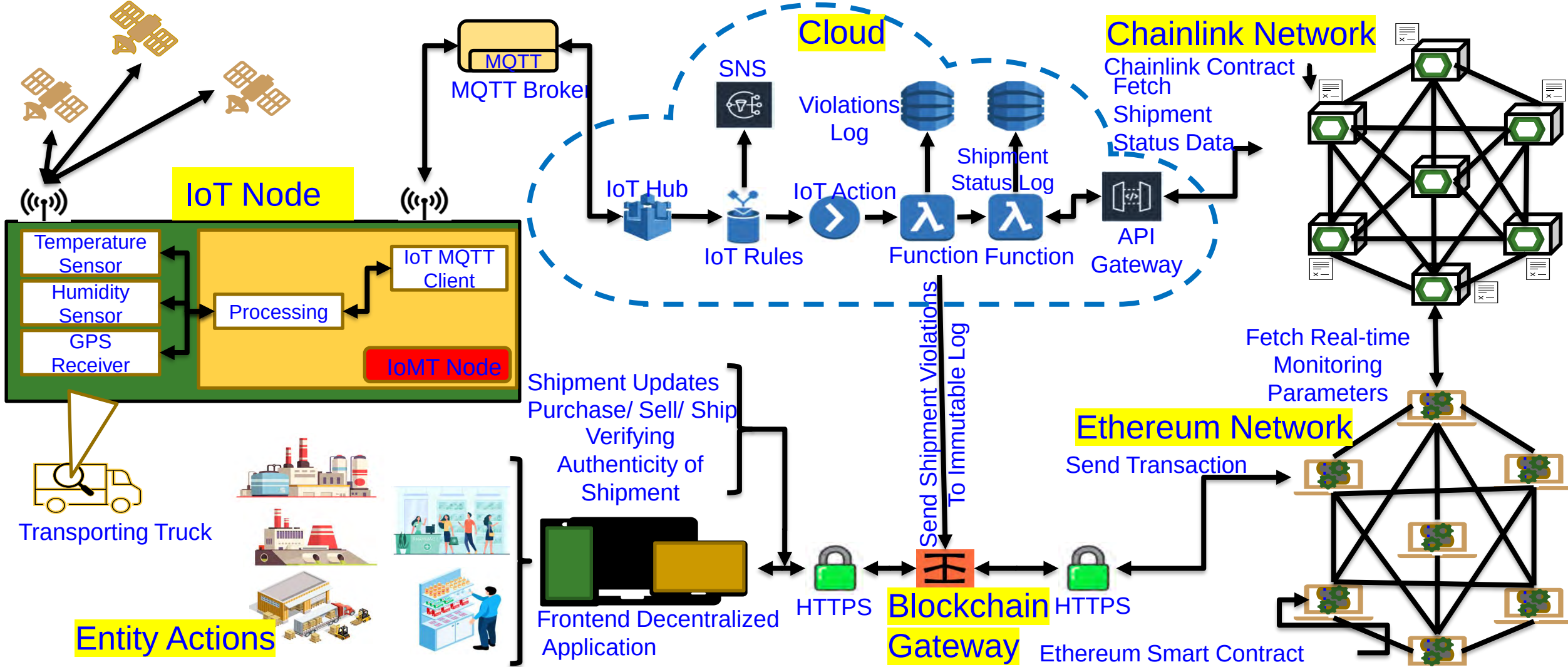
Wholesaler

Consumer

Pharmacy

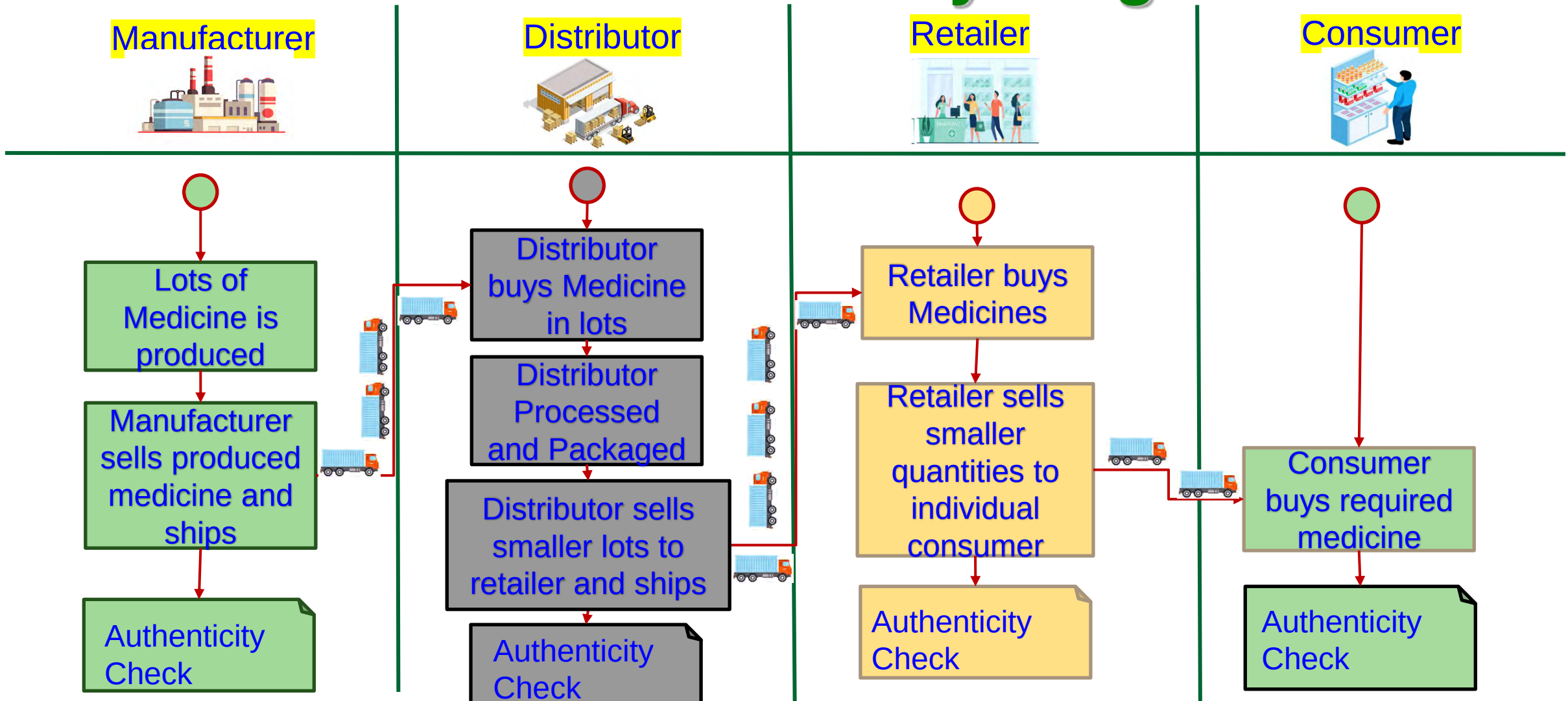
Source: A. K. Bapatla, **S. P. Mohanty**, E. Kougianos, D. Puthal, and A. Bapatla, "PharmaChain: A Blockchain to Ensure Counterfeit-Free Pharmaceutical Supply Chain", *IET Networks*, Vol. XX, No. YY, ZZ 2022, pp. Accepted on 24 June 2022, DOI: <https://doi.org/10.1049/ntw2.12041>. (Dataset for Research: GitHub)

Our PharmaChain: Architectural Overview



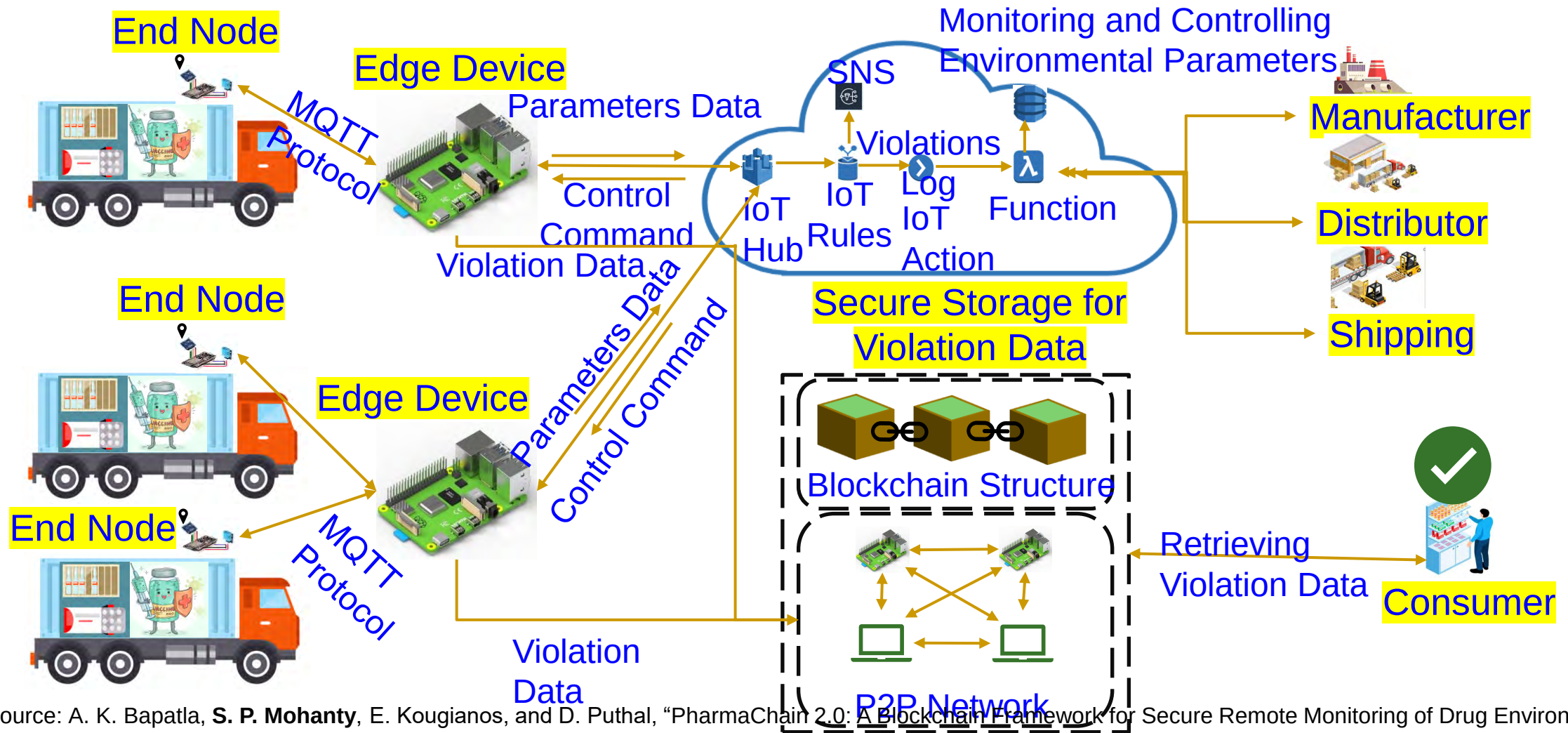
Source: A. K. Bapatla, S. P. Mohanty, E. Kougianos, D. Puthal, and A. Bapatla, "PharmaChain: A Blockchain to Ensure Counterfeit-Free Pharmaceutical Supply Chain", *IET Networks*, Vol. 12, No. 2, March 2023, pp. 53--76, DOI: <https://doi.org/10.1049/ntw2.12041>. (Dataset for Research: [GitHub](#))

PharmaChain Entity Diagram



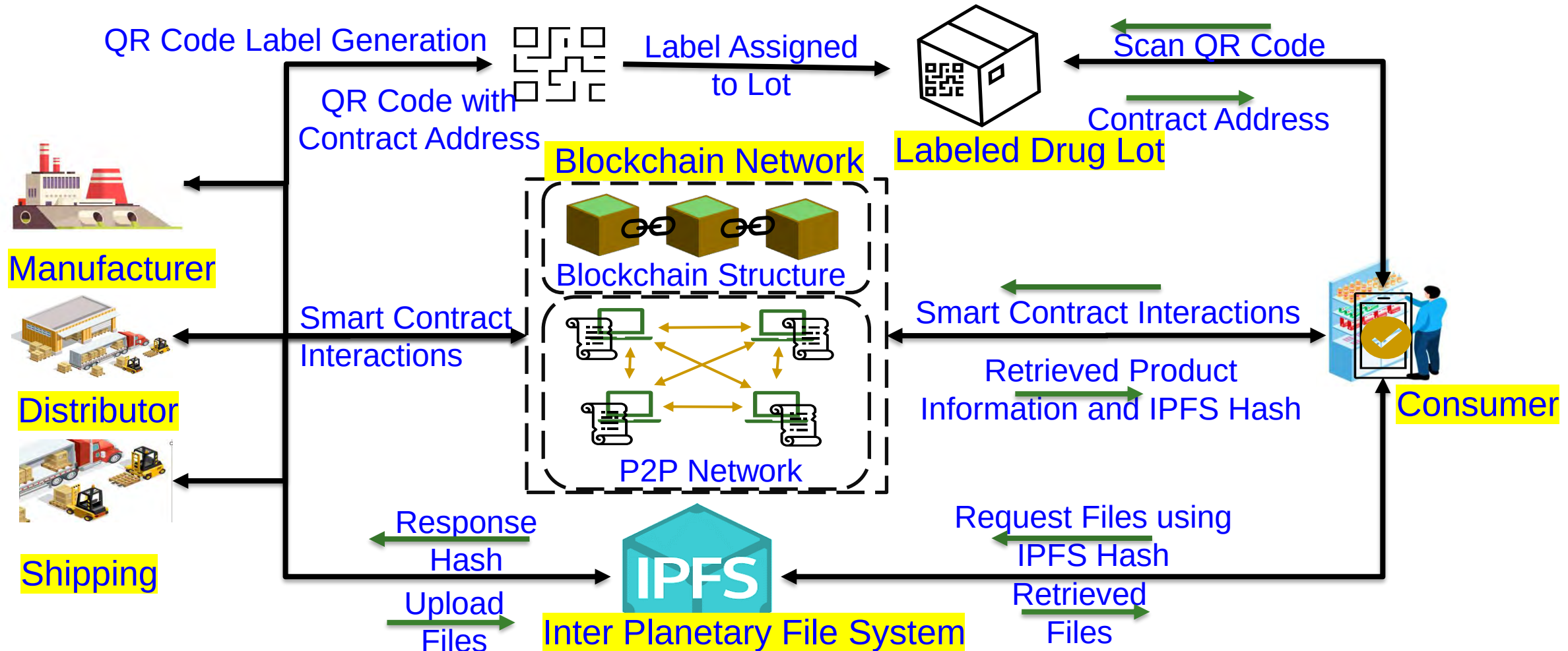
Source: A. K. Bapatla, **S. P. Mohanty**, E. Kougianos, D. Puthal, and A. Bapatla, "PharmaChain: A Blockchain to Ensure Counterfeit-Free Pharmaceutical Supply Chain", *IET Networks*, Vol. 12, No. 2, March 2023, pp. 53--76, DOI: <https://doi.org/10.1049/ntw2.12041>. (Dataset for Research: [GitHub](#))

PharmaChain 2.0 - Architecture Overview



Source: A. K. Bapatla, **S. P. Mohanty**, E. Kougianos, and D. Puthal, "PharmaChain 2.0: A Blockchain Framework for Secure Remote Monitoring of Drug Environmental Parameters in Pharmaceutical Cold Supply Chain", in *Proceedings of the IEEE International Symposium on Smart Electronic Systems (iSES)*, 2022, pp. 185--190, DOI: <https://doi.org/10.1109/iSES54909.2022.00046>.

PharmaChain 3.0 - Architectural Overview

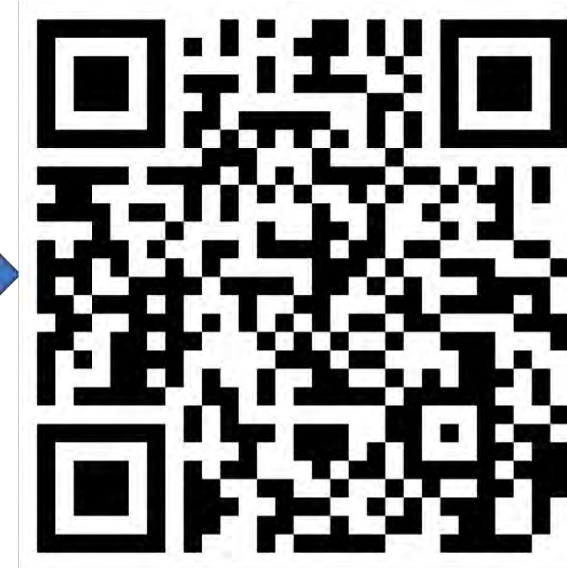


Source: A. K. Bapatla, **S. P. Mohanty**, and E. Kougianos, "PharmaChain 3.0: Efficient Tracking and Tracing of Drugs in Pharmaceutical Supply Chain using Blockchain Integrated Product Serialization Mechanism", *Springer Nature Computer Science (SN-CS)*, Vol. 5, No. 1, Jan 2024, Article: 149, 22-pages, DOI: <https://doi.org/10.1007/s42979-023-02510-9>.

PharmaChain 3.0 – The Key Idea

0x1EcbFd5Edb3747927033
Aa893416e4aD01DF0c6E

Lot Contract Address



QR Code Generated for Lot Labelling

New Lot Contract Address Converted to QR Code for Labeling

Source: A. K. Bapatla, **S. P. Mohanty**, E. Kougianos, and D. Puthal, "PharmaChain 3.0: Blockchain Integrated Efficient QR Code Mechanism for Pharmaceutical Supply Chain", in *Proceedings of the OITS International Conference on Information Technology (OCIT)*, 2022, pp. Accepted.

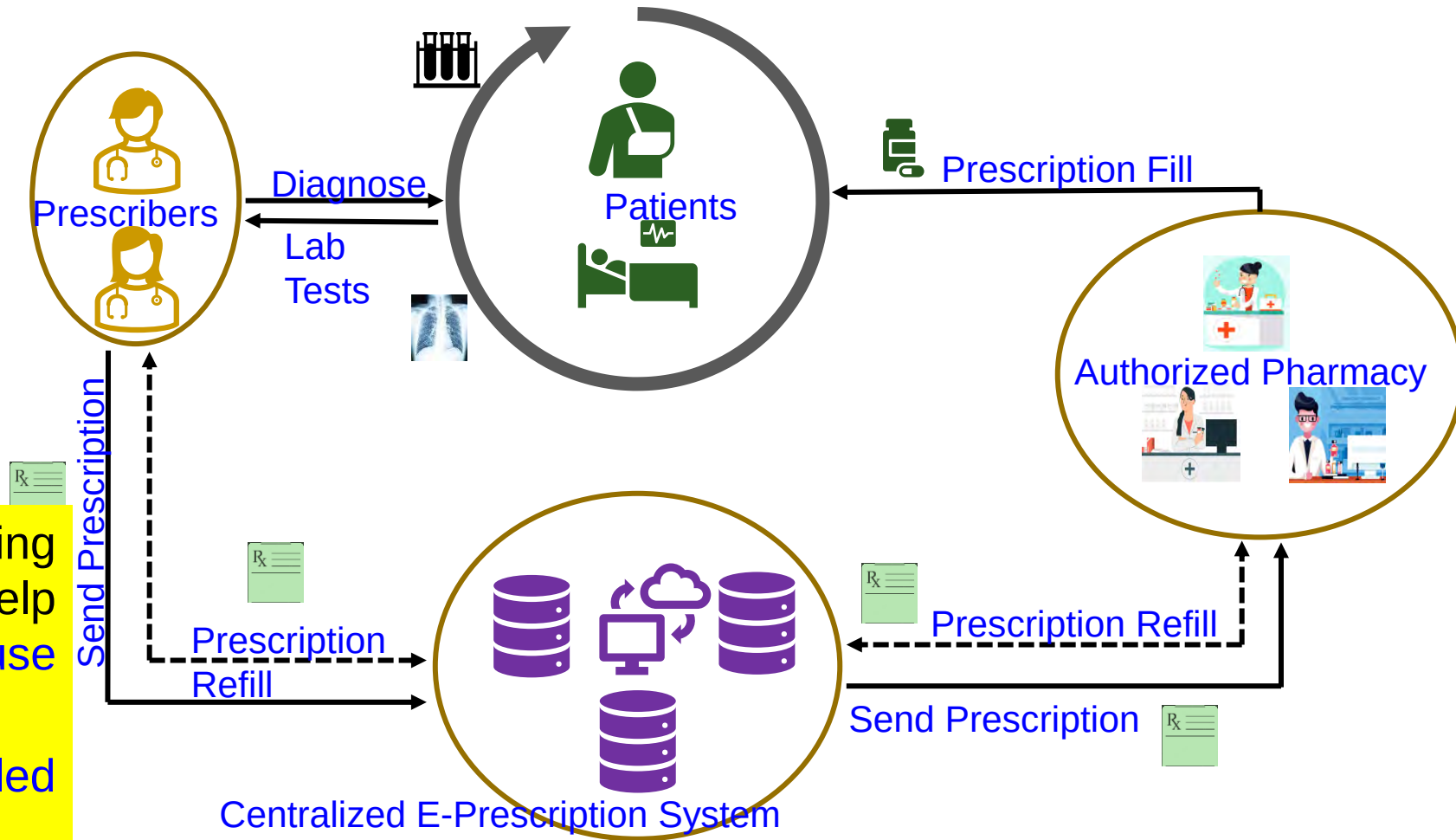
PharmaChain 3.0 - Comparative Analysis

Works	Blockchain	Consensus Mechanism	Computational Needs	Openness	QR Code Integrated	Storage	Handling Large data
Crypto Cargo [11]	Ethereum	Proof-of-Work (PoW)	High	Public	No	On-Chain and Cloud	No
Kumar et.al. [9]	NA	NA	NA	NA	Yes	On-chain	No
PharmaChain [12]	Ethereum	Proof-of-Authority (PoA)	Low	Private	No	On-Chain and Cloud	No
PharmaChain 2.0	Our EasyChain	Proof-of-Authentication (PoAh)	Low	Private	No	On-Chain and Cloud	No
Current Solution (PharmaChain 3.0)	Ethereum	Proof-of-Stake (PoS)	Low	Private	Yes	On-Chain and off-Chain	Yes

Smart Healthcare – Trustworthy Medical Prescription

E-Prescription System and Issues

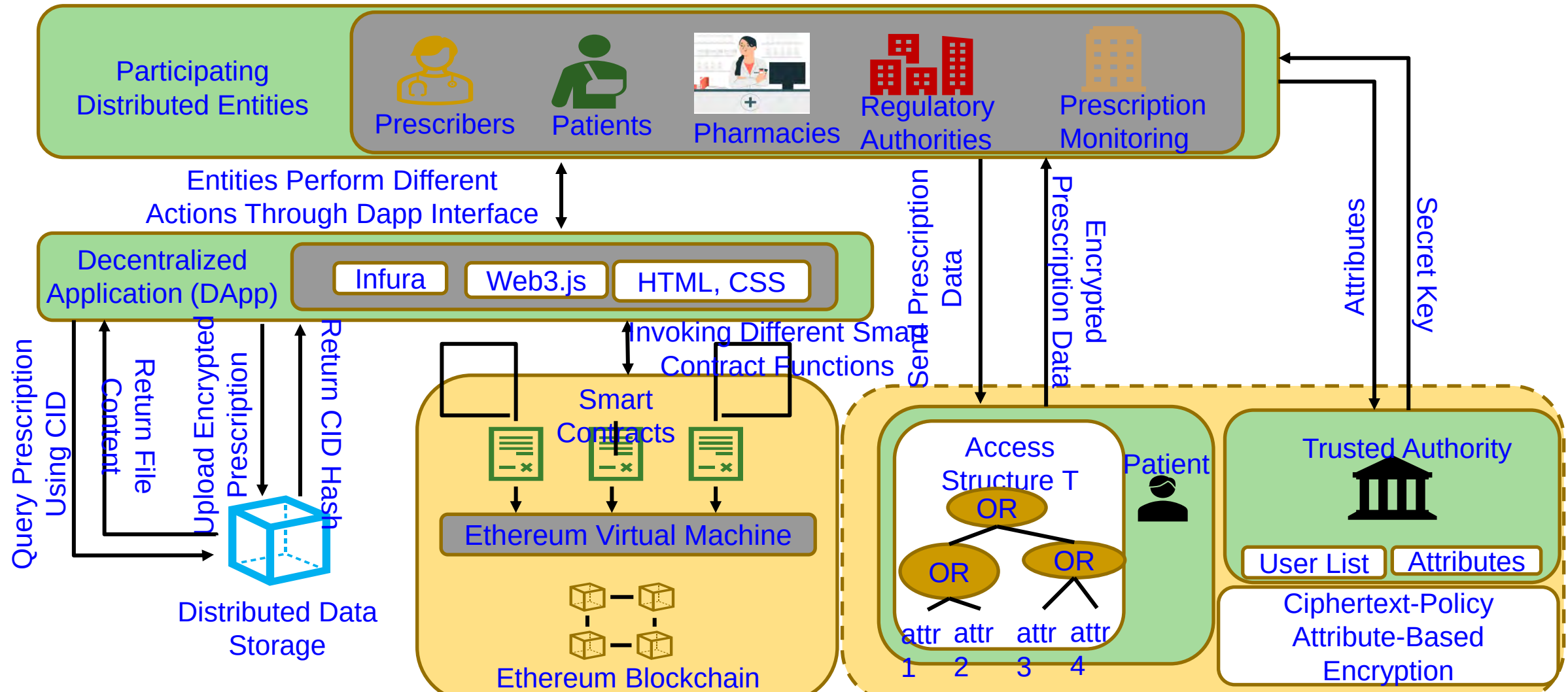
- Single Point of Failure (SPOF)
- Data Security
- Privacy Concerns
- Interoperability Concerns (PDMP)
- System availability Issues



- Prescription Drug Monitoring Programs (PDMP) help mitigate prescription misuse and diversion
- Oversight of controlled substance prescriptions

Source: A. K. Bapatla, **S. P. Mohanty**, and E. Kougianos, "FortiRx: Distributed Ledger Based Verifiable and Trustworthy Electronic Prescription Sharing", in *Proceedings of the IFIP International Internet of Things Conference (IFIP-IoT)*, 2023, pp. 283--301, DOI: https://doi.org/10.1007/978-3-031-45882-8_19.

Our FortiRx: Architecture Overview



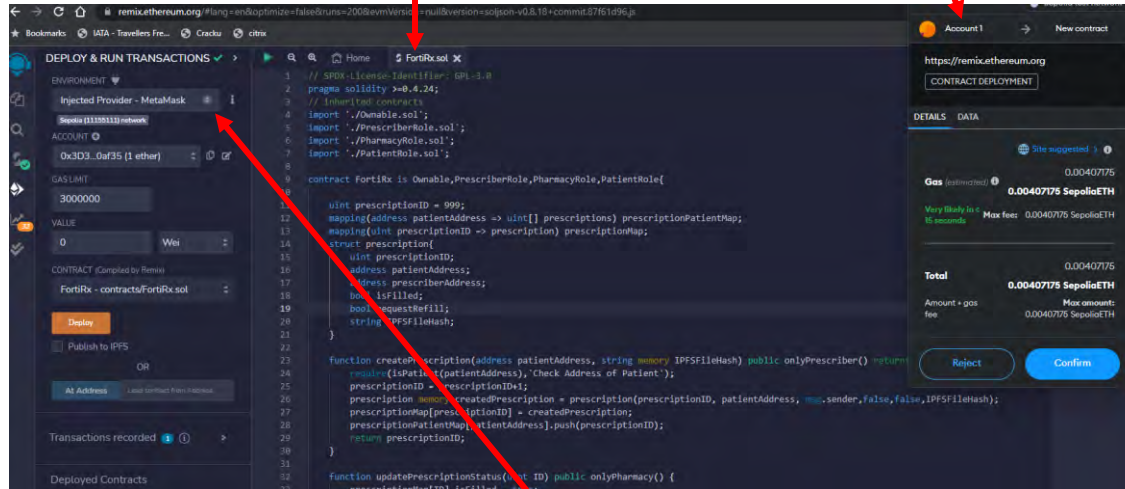
Source: A. K. Bapatla, **S. P. Mohanty**, and E. Kougianos, "FortiRx: Distributed Ledger Based Verifiable and Trustworthy Electronic Prescription Sharing", in *Proceedings of the IFIP International Internet of Things Conference (IFIP-IoT)*, 2023, pp. 283--301, DOI: https://doi.org/10.1007/978-3-031-45882-8_19.

FortiRx: Smart Contract Deployment

Deployment in Sepolia

Smart Contract

Wallet Transaction



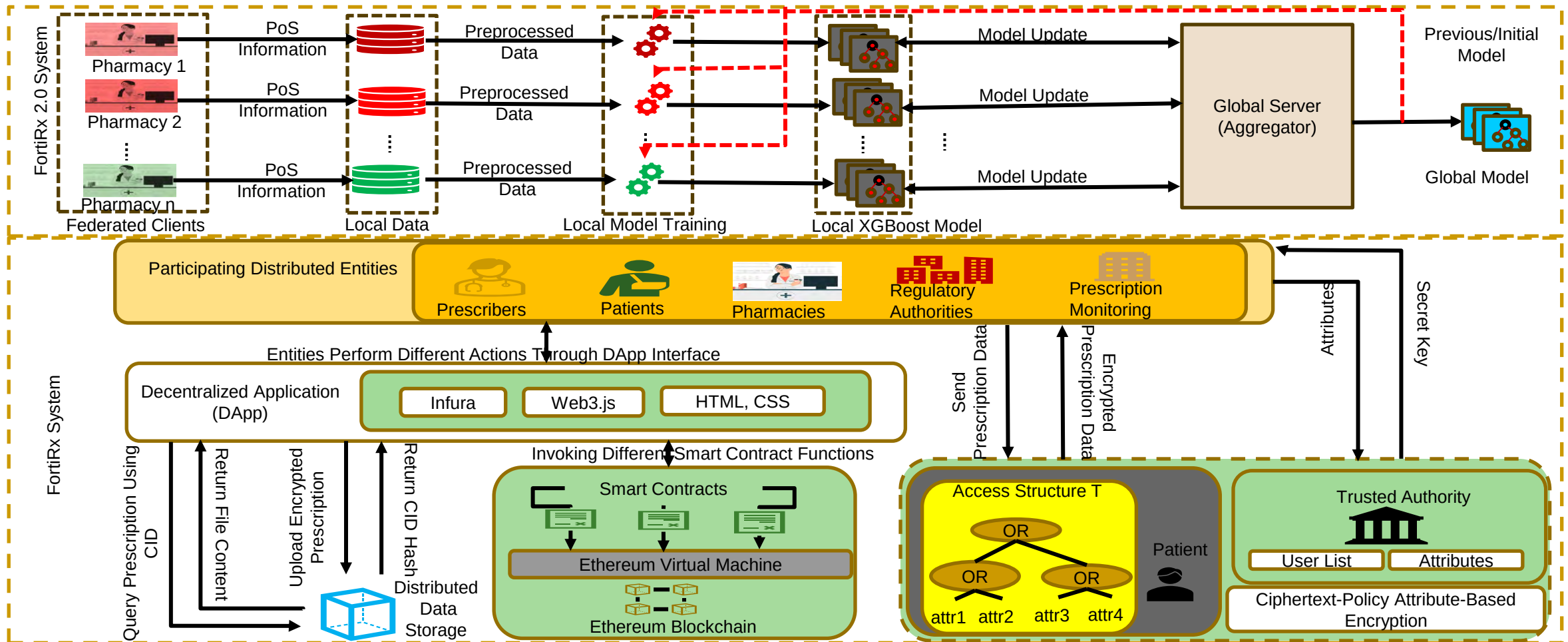
Ethereum Addresses with Roles

Feature	Value
Physician Account Address	0x3d352313f4f5561d0ffbfa205b52a3c3b70af35
Pharmacy Account Address	0x3D352313F4f5561D0ffbfa205B52A3c3b70af35
Patient Account Address	0x2a9884dfa7E6890FE8AA99FE2486c613C32b697a
Contract Deployment Hash	0x798d1f5ff49f9df09b9856db2646cebc2029d5cd2a45c5ef0c1b9acb9f217c6f
Prescription Content ID	Qme7Sq8gLmE875kE79QyWWFy9wqQ4yHnTEHmur511PrZIf
Prescription Creation Hash	0xda5bd0ce943325696e91bfe140bd8cdd60eafdc6f2a41b07221e499bfe7f1f7

Remix Environment Network Configuration

Source: A. K. Bapatla, **S. P. Mohanty**, and E. Kougianos, “FortiRx: Distributed Ledger Based Verifiable and Trustworthy Electronic Prescription Sharing”, in *Proceedings of the IFIP International Internet of Things Conference (IFIP-IoT)*, 2023, pp. 283--301, DOI: https://doi.org/10.1007/978-3-031-45882-8_19.

Our FortiRx 2.0: Architecture



Source: A. K. Bapatla, **S. P. Mohanty**, and E. Kougianos, "FortiRx 2.0: Smart Privacy-Preserved Demand Forecasting of Prescription Drugs in Healthcare-CPS", in *Proceedings of the OITS International Conference on Information Technology (OCIT)*, 2023, pp. 438--443, DOI: <https://doi.org/10.1109/OCIT59427.2023.10430944>.

FortiRx – A Comparative Perspective

Works	Blockchain Platform	Prescription Privacy	Data Management	Drug Demand Forecasting
Ionescu et al, SmartBlock4Health, 2022	Ethereum	Asymmetric Encryption	On-chain	×
VigilRx, 2022	Ethereum	Role-Based Access Control	On-Chain	×
FortiRx, 2023	Ethereum	Role-Based Access Control and CP-ABE	On-chain and off-chain	×
FortiRx 2.0	Ethereum	Role-Based Access Control and CP-ABE	On-chain and off-chain	✓

Source: A. K. Bapatla, **S. P. Mohanty**, and E. Kougianos, “[FortiRx 2.0: Smart Privacy-Preserved Demand Forecasting of Prescription Drugs in Healthcare-CPS](https://doi.org/10.1109/OCIT59427.2023.10430944)”, in *Proceedings of the OITS International Conference on Information Technology (OCIT)*, 2023, pp. 438--443, DOI: <https://doi.org/10.1109/OCIT59427.2023.10430944>.

Conclusion and Future Research



Conclusion

- Healthcare has been evolving to Healthcare-CPS (H-CPS).
- Internet of Medical Things (IoMT) is key for smart healthcare.
- Smart healthcare can reduce cost of healthcare and give more personalized experience to the individual.
- IoMT has advantages but also has limitations in terms of cybersecurity; thus challenging to build sustainable healthcare.
- Cybersecurity in smart healthcare is a serious challenge as device as well as data security and privacy are important.
- Medical device security is a difficult problem due to resource and battery constraints; thus challenge for sustainable H-CPS.
- Security-by-Design is critical for IoMT/H-CPS.

Future Research

- TinyML for smart healthcare that can run at user-end (edge/sensor) needs research.
- H-CPS requires robust data, devices, along with cybersecurity and privacy assurance to be sustainable and hence needs research.
- Security of IWMDs needs to have extremely minimal energy overhead to be useful and hence needs research.
- Integration of blockchain for smart healthcare need research due to energy and computational overheads associated with it.
- SbD research for IoMT/H-CPS is needed.
- Trustworthy Pharmaceutical Supply Chain needs research.